

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

Key Project Information

SECTION A - Description of project

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SECTION B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

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SECTION C - Duration and crediting period

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SECTION D - Summary of Safeguarding Principles and Gender Sensitive Assessment

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SECTION E - Outcome of Stakeholder Consultations

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Appendix 1 - Safeguarding Principles Assessment (mandatory)

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Appendix 2 - Contact information of Project participants (mandatory)

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Appendix 3 - LUF Additional Information (project specific)

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Appendix 3 - Summary of Approved Design Changes (project specific)

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KEY PROJECT INFORMATION

GS ID of Project	GS11169
Title of Project	Aqua Clara Safe Water Program
Time of First Submission Date	28/04/2021
Date of Design Certification	N/A <u>21/04/2022</u>
Version number of the PDD	0809
Completion date of version	0111 <u>0405</u> /2022
Project Developer	Aqua Clara International
Project Representative	ClimateCare Bidco Limited
Project Participants and any communities involved	Aqua Clara International
Host Country (ies)	The Republic of Kenya
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	N/A
Methodology (ies) applied and version number	Methodology for Emission Reductions from Safe Drinking Water Supply V 1.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Total emission reductions	<u>163,656</u> 210,085	tCO2e
SDG 8	Number of people employed by the project	50	Number of people
SDG 6	Number of people attending water hygiene education campaigns.	100	Number of persons
	Number of people attending training/ workshops on filter installation and maintenance.	30	Number of people
SDG 3	Reduction in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases	100%	Percentage
	Improvement in Indoor Air Quality	100%	Percentage

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

The Aqua Clara Safe Water Program hereinafter referred to as the 'project activity', is a program that seeks to distribute point of use water filters to households within 32 Counties in the Republic of Kenya. The project has been developed by Aqua Clara International. The filters technology comprises of the BioSand Filter (BSF) and Hollow fibre filters (HFF) for both households, institutional, and community use. By using the water filters, the project displaces the use of non-renewable biomass fuels such as firewood and charcoal, traditionally used to boil water for drinking and in the alternative offer a cost-effective, sustainable, easy-to-use, and accessible solution to water treatment, especially among households and institutions that consume unsafe raw water.

Additionally, the project activity will actively reduce CO₂ emissions and associated Indoor Air pollution (IAP) associated illnesses among the benefitting end-users. Increased access to safe water for drinking will also significantly contribute towards reduced incidences of waterborne diseases.

The start date of the project is 01/05/2020 which marks the date when the first filter was distributed under the project. The project activity seeks to ensure continuity of the previous water filtration project GS 1078 "Aqua Clara Water Filtration Program in Kenya", by Aqua Clara International whose crediting period will end on 27th December 2021. This is to ensure households within the target counties continue benefiting from the project, and to extend to counties that were not covered by the former project.

However, to ensure there is no incidence of double counting, the following measures will be undertaken:

1. The distribution of filters under the current project was halted to avoid overlap and only the filter sales from 01/05/2020 are registered under the new project. The PP has submitted the most updated sales record for the old project (GS 1078). The sales record is expected to remain the same till the end of the crediting period since no more sales will be added to the project. The summary for the filter sales end dates for the old project and filter sales start date for the new project is as follows:

Old project	Filter sales End date	New project	Filter sales start date
BSF	28/02/2019	BSF	No sales yet
HHFF	30/04/2020	HHFF	01/05/2020
IHFF	28/01/2020	IHFF	27/07/2020
CF	No community filters were sold in the old project.	CF	22/05/2020

2. The PP shall commit to cross check the sales records for the project that is ending and the new project during each verification period to ensure no filter from the old project is added to the new project. This will be verified by cross checking the unique filter ID assigned to each filter, the end-user details and the date of sale.

3. The PP has also revised the crediting period start date to 09/02/2022 which is the start date of design certification or the end date of design review whichever falls latter. to ensure there is no overlap in the crediting periods of the two projects. The crediting period for the old project ended on 27/12/2021.

The WASH joint monitoring programme report (2019) by The World Health Organization and UNICEF found that only 59% of Kenyans have access to basic water services and only 29% have access to sanitary services¹. The report further notes that Kenya has the third-largest number of people in Sub-Saharan Africa estimated at around 9.4 million, who drink water directly from contaminated surface water sources. This makes it essential for the project to continue so that more households can access safe water and other benefits present by the project such as WASH and hygiene campaigns and workshops.

¹ [Water, Sanitation and Hygiene | UNICEF Kenya](#)

The proposed project activity targets to distribute and install at least 500 BioSand Filters (BSF); 77,500 Household Hollow Fibre filters (HHFF); 1,300 Institutional Hollow fibre filters (IHFF) and 120 Community filters within the targeted 32 counties in Kenya. By the time of this submission, the project has distributed 1,806 Household Hollow Fibre Filters, 110 Institutional Hollow Fibre Filters, and 10 community filters within 19 out of the 32 counties which form the project boundary.

Goal and Objectives of the Project Activity

The goal of the project activity is to provide safe water for drinking to households and institutions within the target counties.

Objectives

1. To increase access to water filters which will provide safe water for drinking to households and institutions.
2. To create hygiene awareness among target communities through Water, Sanitation and Hygiene (WASH) sensitisation campaigns and workshops.
3. To improve community livelihoods through employment creation and capacity building.

Implementation plan

The filters will be distributed by Aqua Clara Kenya and their selected partners such as saccos and microfinance institutions. Aqua Clara will regularly train the identified sales agents on how to assemble the filters and conduct basic maintenance.

The target of the project is to distribute 500 BioSand Filters (BSF); 77,500 Household Hollow Fiber filters (HHFF) for household use; 1,300 Institutional Hollow fibre filters (IHFF) and 120 community filters for institutional use. The filters will be distributed within the counties included as part of the project boundary. ~~163,656 210,085~~tCO₂e emission reductions will be produced annually and the total emission reductions during the first crediting period are ~~818,281 1,050,426~~tCO₂e.

Project Boundary

The Aqua Clara water filters are an investment by Aqua Clara International. The project activity shall be implemented in the Republic of Kenya, within the identified project boundary by Aqua Clara Kenya. The project boundary includes households,

institutions, and communities within the 32 identified counties in the Republic of Kenya. These counties include Meru, Tharaka-Nithi, Embu, Kitui, Machakos, Makueni, Nyandarua, Nyeri, Kirinyaga, Murang'a, Kiambu, Trans Nzoia, Uasin Gishu, Elgeyo Marakwet, Nandi, Laikipia, Nakuru, Narok, Kajiado, Kericho, Bomet, Kakamega, Vihiga, Bungoma, Busia, Siaya, Kisumu, Homa Bay, Migori, Kisii, Nyamira, and Nairobi. More details about the project boundary is provided in section A.2 of this PDD.

Baseline scenario

The baseline scenario for this project activity has been identified as provided under section 3.4 of the applicable methodology. The baseline scenario assumes that for users that boil unsafe water for drinking in the pre-project scenario, the general baseline scenario is that users would have boiled water for drinking in the absence of the project activity using non-renewable biomass (firewood and charcoal).

The methodology also provides that for household end-users currently drinking unsafe water, the principles of suppressed demand can be applied, such that the general baseline scenario is assumed to be that users would have boiled water for drinking in the absence of the project activity. However, the suppressed demand baseline does not apply to a large-scale project. A large-scale project can only account for the users that boil water in the pre-project scenario. The suppressed demand baseline may be applied for institutional end-users, except where the institution is connected to a public distribution network (PDN) that supplies safe drinking water - unless justified that supplied water quality doesn't meet safe water definition.

Based on the above and given that the project is large scale, the principle of suppressed demand shall not be applied. More details about the baseline scenario determination is provided in section B.4 of this PDD.

A.1.1. Eligibility of the project under Gold Standard

The project activity seeks to introduce new zero energy Point of Use and Point of Entry water treatment units, namely the BioSand water filters and Hollow fibre membrane filters for households, institutional and community use. By using the filters, the project activity will displace the use of fuelwood traditionally used to treat

water through boiling. The project activity, therefore, qualifies under Sectoral Scope 03 as an Energy Demand activity. The project is eligible under the Gold Standard methodology: Methodology for Emission Reductions from Safe Drinking Water Supply, Version 1.0.

The technologies being promoted by the project activities are eligible under the different technologies outlined in the methodology which include:

- Community level water treatment technologies (CWT)-this covers the community filters (CF) which adopts the use of hollow fibre technology.
- Household water treatment technologies (HWT)-this will include the household BioSand filters (BSF) and the household hollow fibre filters (HHFF).
- Institutional water treatment technologies (IWT): this will include the institutional hollow fibre filters (IHFF) for institutional use.

The proposed project falls under section 3.1.1 of *GS4GG Principles & Requirements* (Version 1.2), section 3 of *GS4GG Community Services Activity Requirements* (Version 1.2) and section 2 of *GS4GG GHG Emissions Reduction & Sequestration Product Requirements* (Version 2.0) with the following eligibility criteria:

Eligibility Criteria	Justification
(a) Types of Project Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section. Section 3.1.1 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2)	The project involves introduction of Biosand and hollow fibre filter technologies which will provide safe drinking water for households and institutions in the 32 counties identified as the project boundary. This will accelerate access to safe drinking water, reduce consumption for woody fuels previously required to treat drinking water and reduce the incidences of waterborne diseases. Therefore, the project activity falls under section 3.1.1 (b) "End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as

Pre-identified CSA project types are a) Renewable energy; b) End-use energy efficiency; c) Waste management and handling; d) Water, sanitation and hygiene (WASH). Section 5.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> The Following Project types are eligible for issuance of Gold Standard VERs: a) Renewable Energy Supply; b) End-Use Energy Efficiency Improvement; c) Waste Handling & Disposal; d) Land Use and Forests.	compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end.” and (d) “Water, sanitation and hygiene (WASH): WASH activities contributing to climate change mitigation and/or adaptation benefits.” of <i>GS4GG Community Services Activity Requirements (Version 1.2)</i>. Besides, the project activity falls under section 5.1.1 (b) “End-Use Energy Efficiency Improvement: Project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end user of the products and services are clearly identified and when the physical intervention is required at the user end.” of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i>. As per section 4.1.3 of <i>GS4GG Principles & Requirements (Version 1.2)</i>, “A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it, or it’s referenced in the Gold
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	Standard Product Requirements". The project type is automatically eligible for Gold Standard Certification.
(b) Location of Project: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects may be located in any part of the world. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country.

(b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	
(c) Project Area, Project Boundary and Scale: Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. To avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. The project boundary is defined based on <i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0) and is limited to the Republic of Kenya (refer to section B.3 of the PDD). The project activity will use zero energy water purification technologies that do not deliver thermal energy. According to section B.4 of the PDD, the baseline scenario of the project is the practice of treating water for drinking by boiling it on stoves using non-renewable biomass. As discussed in section A.4, the annual

double counting or misestimation of impacts amongst projects). Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements. The definition of scale is the same for all Projects, except Microscale. Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or	average thermal energy savings for households is estimated at 2,056.76 GWh_{th} while that for institutions is estimated at 1,053.58 GWh_{th}², which is more than 180 GWh_{th}. As per section 9.1.1 and 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i>, it's a large-scale GS VER project. Each water filter will have a unique serial number for identification. Sales records will be kept for cross-check. The project proponents will check on Gold Standard, CDM, VCS and other voluntary or compliance standard programme based on the repeated serial numbers, similar technology, and project location.
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² 1GJ=3/(3.6*10³)=8.3333*10⁻⁴GWh_{th}

(b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document. Section 9.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> GSVER Projects may be registered as 'large scale', 'small scale' (for the applicability of methodologies and tools only) or 'microscale'. Scale is defined in the relevant Activity Requirements or where these do not exist then per following paragraphs. Section 9.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> All Projects exceeding the small-scale thresholds are defined as large scale. Small scale projects are defined in accordance with CDM project standard for project activities, as below; (a)Type1: Renewable energy Projects: maximum output capacity of 15 MW(e) or 45MW(th). (b) Type 2: End-use energy efficiency project improvement: activities that reduce energy consumption, on the supply and/or demand side, with a maximum energy saving of 60 GWh per year (or an appropriate equivalent) in any year of the crediting period. In this	
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context, for project activities that improve thermal energy efficiency, the maximum energy saving of 60 GWh(e) per year is equivalent to 180 GWh(th) per year saving; (c) Type 3: Other project activities: project involves technologies such Safe Water Supply, Waste management, etc. not included in Type I or Type II that result in GHG emission reductions not exceeding 60,000-ton CO₂e per year in any year of the crediting period.	
(d) Host Country Requirements Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations. Section 3.1.2 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The project location is the Republic of Kenya (refer to section A.2 of the PDD for detailed project area and boundary) and there are no mandatory operational schemes to reduce GHG emissions in any form (e.g., cap & trade, carbon tax etc.) in Kenya, which based on section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) is an eligible host country. A detailed assessment on the project's compliance with applicable Host Country's legal, environmental, ecological, and social regulations is summarized in the table below. The implementation of the project is fully in line with the relevant water resources policies of Kenya. The project technology

Section 3.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> Gold Standard VER Projects may be located in any host country or state. However, where host countries or states have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.), Projects shall only be eligible if the Project Developer has either: (a) provided Gold Standard with satisfactory justification that no double counting of emission reductions occurs or (b) has committed to retiring eligible units equal to the quantity of Gold Standard VERs. Refer to Annex A of this document.	enables users in rural and urban regions to obtain safe drinking water using the Aqua Clara filter technologies without affecting the water source and supply. The implementation of this project can improve users' access to safe drinking water, which is in line with Kenya's vision on water resources. Besides, Kenya submitted its updated Nationally Determined Contribution (NDC) on 24/12/2020, which commits to abate GHG emissions by 32% by 2030 relative to the business as usual (BAU) scenario of 143 million tCO₂e³. The project is in compliance with applicable Host Country's legal, environmental, ecological and social regulations, benefits the local residents and will help in contributing towards Kenya's National emission reductions target. The project activity seeks to ensure continuity of the previous water filtration project "Aqua Clara Water Filtration Program in Kenya (GS 1078)", by Aqua Clara International whose crediting period will end on 27th December 2021. This is to ensure households within the
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³ [https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20\(updated%20version\).pdf](https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Kenya%20First/Kenya's%20First%20%20NDC%20(updated%20version).pdf)

target counties continue benefiting from the project, and to extend to counties that were not covered by the project boundary of the old project.

However, to ensure there is no incidence of double counting, the following measures shall be undertaken:

1. The distribution of filters under the old project was halted to avoid overlap. Only the filter sales from 01/05/2020 (which mark the project start date) are registered under the new project. The PP has submitted the most updated sales record for the old project (GS 1078). The sales record is expected to remain the same till the end of the crediting period since no more sales will be added to the project. The summary for the filter sales end dates for the old project and filter sales start date for the new project is as follows:

Old project	Filter sales End date	New project	Filter sales start date
BSF	28/02/2019	BSF	No sales yet
HHFF	30/04/2020	HHFF	01/05/2020

	IHFF	28/01/2020	IHFF	27/07/2020
	CF	No sales	CF	22/05/2020
	2. The PP will commit to cross check the sales records for the project that is ending and the new project during each verification period to ensure no filter from the old project is added to the new project. This will be verified by cross checking the unique filter ID numbers assigned to each filter and end-user details. 3. The crediting period start date for this project has been set as 09/02/2022 (date of design certification), which is after the crediting period of the ongoing project to ensure there is no overlap in the crediting periods of the two projects.			
(e) Contact Details Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organization (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity	The Project Participants' contact details are provided in Appendix 2 of this PDD.			

registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	
(f) Legal Ownership Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising. Section 3.1.4 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2)	Each water filter end user, who is the default owner of emission reductions, is notified that they waive ownership of emission reductions upon sale of each water purifier. The waiver will be signed with each end-user when selling the water filter to make sure the project owner has the legal ownership of the emission reductions generated by the project activity. Refer to section A.1.2 of the PDD for further details. Moreover, in the stakeholder consultation meeting held on 21/07/2021, the end-users and PP discussed the ownership, and determined that the project owner had the undisputed carbon legal ownership.

Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	
(g) Other Rights Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas.	The project involves installation of water filters to households and institutions within the 32 counties, to provide safe drinking water. The water filtration technologies adopted by the project requires no energy input or consumables. The project does not involve any activity that causes alteration of any resource, or contested legal rights and other disputes, therefore the need for acquiring any specific legal right is not applicable.

(h) Official Development Assistance (ODA) Declaration Section 3.1.1 of <i>GS4GG Principles & Requirements</i> (Version 1.2) All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification. Section 6.1.1 and 6.1.2 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements</i> (Version 2.0) Projects are ineligible for carbon crediting under Gold Standard if the ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly, to the donor country providing ODA support. Project Developer submitting a Project located in a country named by the OECD Development Assistance Committee's	The project owner has signed the ODA declaration and confirms that no ODA has been provided for the project activity.
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ODA recipient list shall sign and submit the ODA Declaration.	
(i) Suppressed Demand Section 3.1.3 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2) Certain Impact Quantification methodologies allow projects to account Suppressed Demand scenario when establishing a baseline. In such cases, the application of Suppressed Demand baseline is limited to Small Scale and Microscale Projects. Where a Suppressed Demand baseline is applied, it is not possible to 'stack' Gold Standard Certified Impact Statements or Products as the definition of the baseline may be contradictory.	The project activity will claim credits for end-users that boil water. The same has been confirmed through the baseline survey conducted. As per Section 3.1.3 of <i>GS4GG Community Services Activity Requirements</i> (Version 1.2), the project activity is a large-scale GS VER project.
(j) Eligible Greenhouse Gases Section 4.1.1 of <i>GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0)</i> Only Carbon Dioxide (CO₂), Methane (CH₄) and/or Nitrous Oxide (N₂O) are eligible for Gold Standard crediting, provided Projects comply with Gold Standard Requirements and eligibility criteria.	The project activity only considers the reduction of Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O) Emissions for Gold Standard crediting. Refer to section B.3 of the PDD for further details.

Eligibility to Community Service Activity Requirements

The project activity is a water filtration project and is eligible under End-Use Energy efficiency projects. The use of the water filters shall displace the demand for non-renewable biomass (firewood and charcoal) which is used to boil drinking water in the baseline scenario.

Gold Standard for the Global Goals Applied Eligibility Principles

Principle 1: Contribution to Climate Security & Sustainable Development:

The project activity will contribute positively to Climate Security & Sustainable Development. This has been assessed under the Sustainable Development Goals in section B.6 of the PDD.

Principle 2: Safeguarding Principles:

The project has conducted a Safeguarding Principles Assessment and has ensured conformance to the Gold Standard Safeguarding Principles and Requirements. The assessment is included in Appendix 1 of the PDD.

Principle 3: Stakeholder Inclusivity:

The project developer conducted a stakeholder consultation meeting on 21/07/2021, in accordance with the Gold Standard Stakeholder Consultation & Engagement Procedure, Requirements & Guidelines. The stakeholder feedback has been considered in the design, planning, and implementation of the Project. Additionally, the project will continue soliciting the views and inputs of stakeholders through the continuous input and grievance procedure to ensure all issues are captured and acted upon throughout the lifetime of the Project.

Principle 4: Demonstration of real outcomes:

The project shall demonstrate real outcomes by undertaking the following:

- Designing and developing a Project Design Document (PDD), that incorporates a Monitoring & Reporting Plan.
- A design certification that shall comprise of Validation and Design Review
- Monitoring in accordance with the Monitoring & Reporting Plan
- Performance Certification comprising of Verification and Performance Review

Principle 5: Financial Additionality & Ongoing Financial Need:

The project has demonstrated additionality as discussed in section B.5 of the PDD.

Registration with other Carbon standards

The project activity is not registered with any other carbon standard, nor is it seeking registration under any other GHG programs, and hence there is no likelihood that double counting would occur. Additionally, each filter shall be provided with a unique filter serial number which gives a distinction to the end users which will further ensure there is no double counting.

Compliance with applicable Host Country's legal, environmental, ecological, and social regulations.

Applicable regulation	Provisions of the law	Project's compliance
2010 Kenya's Constitution	This is the principal guiding law in the country from which all the subsidiary laws are drawn from. It guarantees and gives every person in the country the right to enjoy a safe and clean environment.	The project activity seeks to provide clean and safe drinking water which will benefit the end-users. This will lead to improved health due to reduced waterborne diseases and indoor air pollution and promote hygiene and sanitation which is paramount in ensuring a clean and safe environment and human wellbeing.
Climate Change Act, 2016 ⁴	The Climate Change Act (No. 11 of 2016) is the first comprehensive legal framework for climate change governance for Kenya. The objective of the Act is to "Enhance climate change resilience and low carbon development for sustainable development of Kenya."	The project activity is promoting the use of zero energy, point of use water filtration technologies. By using the water filters, the pre-project practice of boiling water for drinking using non-renewable biomass is displaced. Through this, the project is contributing towards mitigating climate change in Kenya.

⁴ kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/ClimateChangeActNo11of2016.pdf

Environmental Management and Coordination Act, 1999 ⁵ (2015 amendment)	This is an Act of parliament to provide for the establishment of an appropriate legal and institutional framework for the management of the environment and for matters connected therewith and incidental thereto. Part VII on Environmental Audit and Monitoring section 58 specifically detail the need to undertake Environmental Impact Assessment of all projects likely to cause negative impacts to the environment as listed in the second schedule of the act. Further, part V of the Environmental Impact Assessment and therefore mandatory that an Environment Impact Assessment process.	Water filtration projects are not listed in the second schedule and therefore no EIA is required for the project activity. The project is anticipated to have numerous positive environmental and social impacts.
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⁵ [ken41653.pdf \(fao.org\)](#)

Employment Act, 2007 ⁶ .	The Act declares and defines the fundamental rights of employees, to provide basic conditions of employment of employees, to regulate employment of children, and to provide for matters connected with the foregoing. The act provides that an employer shall promote equal opportunity in employment and strive to eliminate discrimination in any employment policy or practice. Further, an employer shall pay his employees equal remuneration for work of equal value.	The project activity shall ensure compliance to this act by providing equal employment opportunities to all without discrimination. Further, the project shall monitor the number of people employed and their remuneration to ensure equal pay for equal value of work done is paid.
Public Health Act, 2012 ⁷	Part XI of the act assigns local authorities a) the responsibility for preventing any pollution dangerous to health of any supply of water which the public	The project activity seeks to complement the efforts being undertaken by local authorities in ensuring households have access to affordable, convenient, and reliable solutions to water

⁶ Microsoft Word - Paged_Employment Act _Cap. 226 - No. 11 of 2007_.doc (kenyalaw.org)

⁷ kenyalaw.org/kl/fileadmin/pdfdownloads/Acts/PublicHealthActCap242.pdf

	within its district has a right to use and does use for drinking or domestic purposes; and (b) for purifying any such supply which has become so polluted, and to take measures (including, if necessary, proceedings at law) against any person so polluting any such supply or polluting any stream so as to be a nuisance or danger to health.	purification. The project will work closely with public health officials within the target areas to ensure they create synergies in enhancing access to safe water.
Standards Act, 2012 ⁸	The purpose of the act is to promote the standardization of the specification of commodities, and to provide for the standardisation of commodities and codes of practice; and to establish a Kenya Bureau of Standards (KEBS).	The project activity has complied with the provisions of the act by undertaking the required tests on the filters and has obtained the KEBS standardization mark which has been provided as supporting evidence.

⁸ [Microsoft Word - Paged Standards Act Cap. 496 - No. 17 of 1973 .doc \(kenyalaw.org\)](#)

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

The legal owner of the project is Aqua Clara International. The VERs generated by the project activity shall be the sole ownership of the Project Participant. During the sale/installation of the units, the end-user signs the sales receipt which includes a statement that the rights to the carbon credits have been relinquished to the project developer. By signing the waiver, the rights to the carbon credits generated by the project are transferred to Aqua Clara and they remain uncontested.

The VER transfer procedure was discussed with the stakeholders during the LSC meetings. This ensured that all stakeholders were made aware that the sole ownership of the carbon credits generated by the project activity belong to Aqua Clara (the project developer). An explanation of how the carbon finance will be utilized was also explicitly explained during the meeting.

Aqua Clara International is a non-profit private organization that exists to aid in the establishment of sustainable locally owned businesses that provide affordable, clean, and safe water for children and families worldwide. The company freely transfers knowledge related to the construction of the water filters from locally available materials as well as in supporting the operation of local businesses that can meet the needs of their communities in a financially sustainable manner.

A.2 Location of project

The project activity will be implemented in 32 Counties within the Republic of Kenya which will form the project boundary. The 32 counties are represented by the co-ordinates below (Source: <https://simplemaps.com/data/ke-cities>):

County	Latitude	Longitude	County	Latitude	Longitude
1. Meru	0.0500	37.6500	2. Laikipia	0.4417	36.7440
3. Tharaka-Nithi	-0.29557	38.015609	4. Nakuru	-0.2833	36.0667
5. Embu	-0.5333	37.4500	6. Narok	-1.0833	35.8667
7. Kitui	-1.3667	38.0167	8. Kajiado	-1.8500	36.7833
9. Machakos	-1.5167	37.2667	10. Kericho	-0.3692	35.2839
11. Makueni	-1.8086	37.4697	12. Bomet	-0.7105	35.2483
13. Nyandarua	-0.2464	36.3337	14. Kakamega	0.2833	34.7500
15. Nyeri	-0.4167	36.9500	16. Vihiga	0.0660	34.6462
17. Kirinyaga	-0.5072	37.2740	18. Bungoma	0.5666	34.5666
19. Murang'a	-0.7210	37.1526	20. Busia	0.4608	34.1108
21. Kiambu	-1.1714	36.8356	22. Siaya	0.0607	34.2881
23. Trans-Nzoia	1.1151	34.8995	24. Kisumu	-0.1000	34.7500
25. Uasin Gishu	0.2663	35.3371	26. Homa Bay	-0.5167	34.4500
27. Elgeyo Marakwet	1.0833	35.4698	28. Migori	-1.0634	34.4731
29. Nandi	0.5088	35.0333	30. Kisii	-0.6817	34.7667
31. Nairobi	-1.2864	36.8172	32. Nyamira	-0.5633	34.9358

TEMPLATE

The Administrative map of the Republic of Kenya

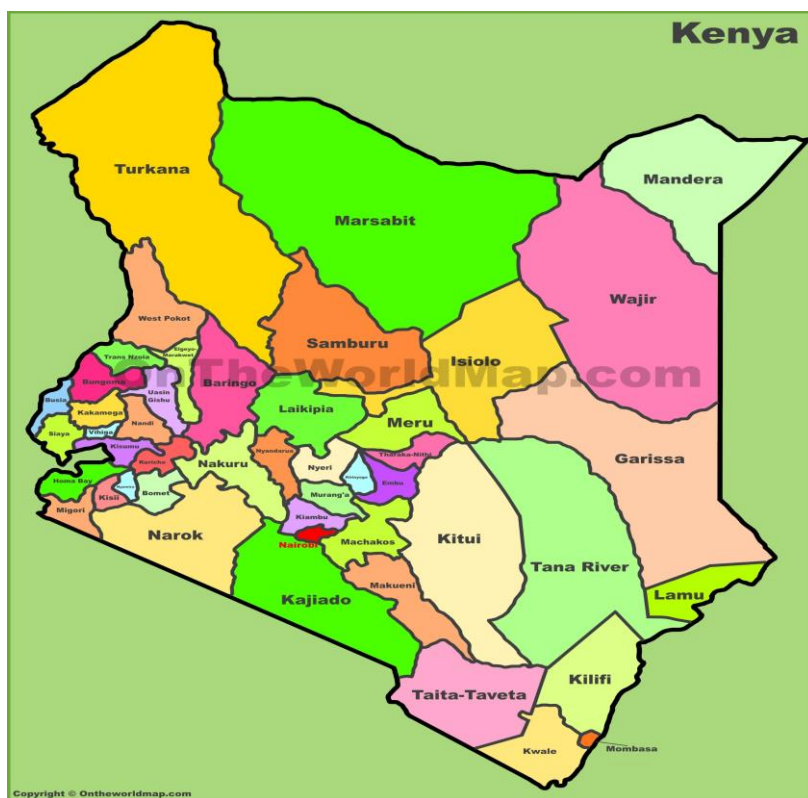


Figure 1: The Administrative Map of Kenya indicating the 47 Counties.

Source: [Administrative map of Kenya \(Counties of Kenya\) \(ontheworldmap.com\)](http://ontheworldmap.com)

Map highlighting the 32 counties included in the project boundary.

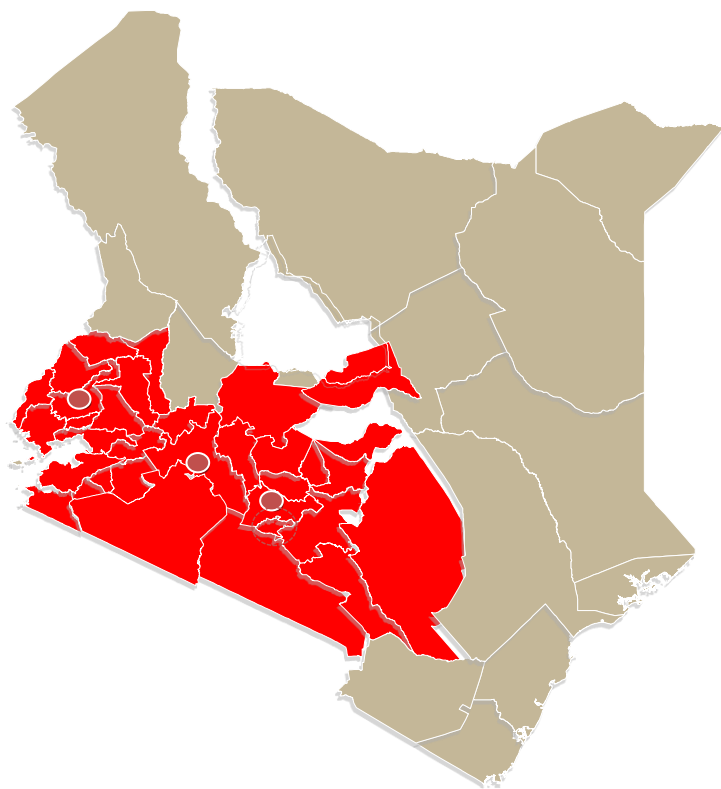


Figure 2: Map highlighting the 32 counties that form the project boundary.

A.3 Technologies and/or measures

Project Technology

This project activity is applicable to two project technologies which include the BioSand filter (BSF) and Hollow Fibre membrane filter units.

1. The Bio Sand Filter (BSF)

The BioSand filters (BSF) are an improvement on the traditional slow sand filters. The technology works using fine sand which filters away larger components such as turbidity, protozoa, and larger bacteria while the biological layer feeds on smaller disease-causing organisms in the water.

The filter is made up of durable food grade 70+ L container and plastic pipes, sand, and ballast. The filters are made in Kenya using 100% locally available materials, providing an inexpensive way to produce high-quality drinking water at the peri-urban and rural household level from almost any water source. The filter has an output of 1litre per minute and can produce up to 60 litres of safe water in an hour.

Maintenance of the ACI BSF is relatively simple as it only requires that the consumer cleans the filter at least once or twice a year depending on the turbidity of the water. Water with high turbidity tend to clog the filter hence reducing the flow rate. However, the user can increase the flow rate by simply adding 10 litres of water to the BSF unit, swirling the top 4-6 cm of sand up into the water with a stick, taking all sediments into suspension, pausing to allow sand grains to settle out and then scooping and dumping the water out. This swirl and dump procedure is easy to conduct and takes only up to 2 hours for the household to access clean water again. This guarantees the consumer continued filtration throughout the lifespan of the BSF unit of 10 years with proper use and maintenance. The first maintenance check is done by the installer of the BSF unit, who has received training from ACI, followed by a demonstration to households on how to go about the maintenance.



Figure 3: The Aqua Clara International BioSand water filter

Protozoa and parasites are 100% eliminated, bacteria are reduced by over 95%, and viruses are reduced up to 70%. Sediments are also eliminated. Pathogens and suspended solids are removed through a combination of biological and physical processes that take place in the bio-layer and within the sand layer.

These processes include:

1. Mechanical trapping where suspended solids and pathogens are physically trapped in the spaces between the sand grains.
2. Predation where pathogens are consumed by other microorganisms in the bio layer.
3. Adsorption where pathogens become attached to each other, suspended solids in the water and the sand grains.
4. Natural death where pathogens finish their life cycle or die because there is not enough food or oxygen for them to survive.

These different processes occur during different points in the filter running process. These can be broken down into segments of the filter running process: the pause period (idle time) and actual run of the filter.

The actual run occurs when the high-water level in the inlet reservoir zone pushes the water through the diffuser and filter. The water level in the reservoir decreases as it flows evenly through the sand. The flow rate will slow down over time because there is less pressure to force the water through the filter. The inlet water contains dissolved oxygen, nutrients, and contaminants. It provides the oxygen required by the microorganisms in the biofilm. Larger suspended particles and pathogens are trapped in the top of the sand, and they partially plug the pore spaces between the sand grains. This also causes the flow rate to slow down. The pause period or idle time typically accounts for greater than 80% of the daily cycle whereupon various microbial attenuation processes are likely to be significant. In other words, most removal occurs during the point where water is in contact with the biofilm. When the standing water layer reaches the same height as the outlet tube, the flow stops, ideally, this should be at a level that is high enough to keep the biofilm in the sand layer wet, but also allow oxygen from the air to diffuse through the standing water to the bio-layer. The pause period allows time for microorganisms in the bio layer to consume the pathogens and nutrients in the water. The flow rate through the filter is restored as they are consumed. If the pause period is too long, the bio-layer will eventually consume all the pathogens and nutrients and die off. This will reduce the removal efficiency of the filter when it is used again. The pause period should be a minimum of 1 hour after the water has stopped flowing up to a maximum of 48 hours. Pathogens in the non-biological zone die off due to the lack of nutrients and oxygen.

The main benefits of the BSF filter are:

- Least expensive filter on the market
- Lasts 10 years+ with proper maintenance
- No boiling of filtered water is required.
- No electricity is necessary
- Can filter river, lake, and borehole water.
- Dirty (turbid) water will not damage the filter.
- Simple maintenance several times a year
- Made up of 100% local materials.

2. ACI Hollow Fibre Membrane Filter)

The ACI Hollow Fibre membrane filter operates using a hollow-fibre membrane technology which for decades, has been used in medical, water, food, and industrial applications for high-precision ultra-filtration. Hollow micro-tubes are created of polymeric material and resembles straight lengths of noodles that are hollow on the inside. The walls of the tubes are permeated with tiny, microscopic pores not greater than 0.1 microns in diameter. Water enters the filter elements on the outside of the tubes. Since bacteria are larger than 0.1 micron, they cannot flow through the microscopic pores. Only water can get through the holes. Pathogens such as bacteria, algae, cysts and parasites, and debris are left on the outside of the tubes. Pure water is collected as it flows on through the filter elements. The filters are easily and quickly backflushed to wash away the impurities that collect in the filter elements. The filtration process is completely mechanical. No biological or chemical processes are involved.

These filters are designed to be used in both in domestic set ups (households, apartments, etc), institutions such as schools, and restaurants as well as commercial facilities such as restaurants and hotels.

The filters were designed and engineered by Aqua Clara in their laboratories at the Bio Economic Institute of Michigan State University, in Holland, Michigan, USA.

Specifications and performance

- Capacity: can produce 100-400litres of water per day.
- High flow rate: 0.3-1.0 litre per minute
- Easy installation: 1 hour training in hygiene and maintenance; then pick up and carry home.
- Start-up: no start-up process required; filter can be used immediately.
- Pause time: no pause time is required between fill-ups.
- Portability: full portability without disrupting filter processes.
- Easy to use and maintain.
- Up to 5 Years lifespan with proper use and maintenance
- Have several models to fit specific needs: designed for homes, apartments, schools, offices, and restaurants.



Figure 4: The Aqua Clara International Hollow Fibre filters, HF15 and HF10 models

School filters

ACI have designed hollow fibre filters specifically for schools. Many schools struggle to provide adequate safe drinking water to all students throughout the school day. The Aqua Clara school filter is an affordable, effective, and lasting solution to this problem. Water from any source can be made safe in a few minutes with sufficient storage to ensure that it is always available when needed.

Extra benefits:

- Has a storage capacity of 100 litres.
- Has 3 taps for dispensing water.
- Comes with own stand, 24 cups and 3 handwashing facilities.



Figure 5: The ACI Hollow Fibre school filter model

ACI Hollow fibre facility filters

Hollow fibre membranes are made from polymeric material and resemble straight lengths of noodles that are hollow on the inside. On the surface are very tiny tightly controlled microscopic pores (0.2 micron) that can stop the passage of particles, turbidity, and pathogens such as bacteria, algae, cysts, and parasites but allow water to flow through. The filter eliminates up to 99.99% bacteria and protozoal parasites.

The ACI Facility Hollow Fibre Membrane Filter unit are designed to purify and supply water to large institutions such as schools, hotels, and clinics, or can be purchased by end-users for use in water kiosks (commercial facilities). All purchases of filtered water from water kiosks shall be tracked per consumer and recorded electronically by the project developer. The average capacity of the ACI Facility Hollow Fibre Membrane Filter shall be about 1,000 to 10,000 litres per day.

ACI Facility Hollow Fibre Membrane Filters features summarily will include:

1. Uses the hollow fibre filtration technology.
2. Uses a sand pre-filter to reduce turbidity.

3. Operates by gravity fed system, no electricity is required.
4. Have an output of up to 24 litres per minute.
5. Have a capacity to produce 1,000 –10,000 litres of water per day.
6. Removes up to 99.99% of bacteria.
7. Easy installation and simple maintenance are required.
8. Water meters fitted to measure water output from the filter.
9. Have a lifespan of 5+ years with proper use and maintenance.



Figure 6: The ACI Facility Hollow Fibre Membrane Filter

A.4 Scale of the project

As per section 9.1.1 and 9.1.2 of GS4GG GHG Emissions Reduction & Sequestration Product Requirements (Version 2.0), the project is a large-scale project. As detailed in section A1.1 of this PDD, the annual average thermal energy savings from the household filters ~~1,622,242,056.76~~ GWh_{th}, while the savings from institutional and community filters is ~~771,371,053.58~~ GWh_{th} which is more than 180 GWh_{th}. The calculations are provided in the Ex-ante emission reduction spreadsheet tab thermal savings and is as summarized below:

Default amount of energy required to boil 1 L of water from a first principles approach (kJ/L)	360.83
Efficiency of the baseline water boiling. Weighted average of baseline stove types (η_{wb})	11.14.2%
Annual average thermal energy savings (GJ)	1,946,683.212,468,116.223
Unit conversion from GJ to GWh _{th}	8.333E-04
Annual average thermal energy savings (GWh _{th})	1,622.242,056.76

Total Volume of water treated (HHHF): 761,181,265.63l⁹

Total Volume of water treated (BSF): 4,910,846.88l¹⁰

As per section 3.6.2 of the applied GS methodology, the amount of energy required to boil 1 L of water is 360.8311 kJ/L. Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling in households is ~~11.214.2%~~.

Annual average thermal energy savings is calculated as follows:

$$= (360.83\text{kJ/L} \times 761,181,265.63 \text{ L} / \del{11.214.2\%} / 1,000,000) + (360.83\text{kJ/L} \times 4,910,846.88\text{L} / \del{11.214.2\%} / 1,000,000) = \del{1,946,683.2182,468,948.9216\text{GJ}}$$

Converted to GWh_{th} = ~~1,622.242,056.76~~ GWh_{th}

Annual average thermal energy savings calculations for institutions and community filters (GWh_{th})

Total Volume of water treated (IHFF): 270,660,000.00l

Total Volume of water treated (CF): 160,314,000.00l

⁹ Assuming 77,500 filters as projected and using a Qy value of 27l

¹⁰ Assuming 500 filters as projected and using a Qy value of 27l

¹¹ This is calculated from the specific heat of water of 4.186 kJ/L °C, the difference between the initial and final water temperature assuming a start at 20 °C and end at 100 °C, evaporation of 1% of water during 5 minutes of boiling to obtain 1 L boiled water, and latent heat of water evaporation of 2260 kJ/L.

Default amount of energy required to boil 1 L of water from a first principles approach (kJ/L)	360.83
Efficiency of the baseline water boiling. Weighted average of baseline stove types (η_{wb})	12.3 16.8%
Annual average thermal energy savings (GJ)	925,644.93 11,264,295.516
Unit conversion from GJ to GWh _{th}	8.333E-04
Annual average thermal energy savings (GWh _{th})	771.37 1,053.58

~~Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling in households is 12.3%.~~

~~Annual average thermal energy savings is calculated as follows:~~

$$\begin{aligned} &= (360.83 \text{ kJ/L} * 761,270,660,000.00 \text{ L} / 12.3\% / 1,000,000) + (360.83 \text{ kJ/L} * 160,314,000.00 \text{ L} / 12.3\% / 1,000,000) = 1,264,295.516 \text{ GJ} \end{aligned}$$

~~Converted to GWh_{th} = 1,053.58 GWh_{th}~~

A.5 Funding sources of project

The project activity has not received, nor will it receive any public funding. To affirm the same, an ODA declaration has been submitted to the Gold Standard as a supporting document.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The following approved Gold Standard baseline and monitoring methodology is applicable to the project activity: Methodology for Emission Reductions from Safe Drinking Water Supply v 1.0.

B.2. Applicability of methodology (ies)

This project qualifies under the above stated methodology which provides the following:

Eligibility Criteria	Justification for the Project Activity
Eligible household water treatment technologies (HWT), institutional water treatment technologies (IWT), and community level water treatment technologies (CWT) include bleach/chlorine, water filter (ceramic, sand, composite, membrane, etc.), UV disinfection, etc.	The project activity will introduce zero emission point of use and point of entry water purification technology which will displace the use of fuel wood traditionally used to treat drinking water. The technologies included in the project include the household water treatment technologies (BioSand filters and Household hollow fibre filters), institutional water treatment technologies (Institutional hollow fibre filters) and community level water treatment technologies (Community hollow fibre filters).
This methodology allows for project activities to include safe water treatment and/or supply technologies implemented for end-users in households, and/or commercial premises such as shops or institutional premises including half or full	The project activity will distribute both household water treatment technologies (BSF and HHFF filters), institutional water treatment technologies (IHFF), and Community level water treatment technologies (CF). The applicable calculations will be used to quantify emissions from the technologies.

day/boarding schools, prisons, army camps & refugee camps.	
All projects involving CWT and CWS technologies must also include ongoing maintenance and repair of the project technology.	CWT technologies are included in the project and the required maintenance and repair shall be undertaken by the project developer where necessary. The Aqua Clara Filter Community water filters use the hollow fiber filtration technology and are maintained through a simple process called backwashing. The backwashing of the filter is done twice a week following the procedure outlined in the Aqua Clara CF filter manual provided as supporting evidence. The maintenance procedure is provided to the recipients of the CF filters, and they are also trained how to do the maintenance during the installation of the filter. The maintenance records will be provided during the monitoring of the project as required.
In cases where the safe water is retrieved at the CWT or CWS location, the water in its improved form shall be available within a distance of 1 km or less from the end-users, as demonstrated by satellite imaging or GPS coordinates of each CWT or CWS location. Alternatively, as a proxy, a total collection time of 30 minutes or less for a round trip, including queuing, using the travel modes of	For the CWT technology, the PD shall ensure the methodological requirement of 1 km distance is met. This will be cross-checked during the monitoring of the project.

walking or pedalling may be demonstrated.	
Project technology performance level (HWT and IWT): It shall be demonstrated based on report of laboratory testing or official notification that the project technology or equipment achieves either (i) the performance target classification 3-star or 2-star level, meaning "Comprehensive Protection," as per the WHO International Scheme to Evaluate Household Water Treatment Technologies (World Health Organization, 2011) or (ii) compliance with the national standard or guideline for household drinking water treatment technology; if no national guideline or standard is available, then the project technology shall comply with the WHO International Scheme requirements.	The technology performance level for household and institutional filters shall be demonstrated following method (ii) to ensure compliance with the Kenya's national standard ¹² and guideline for household drinking water treatment technology. This has been demonstrated under section A.1 of the PDD and the required supporting evidence have been provided.
Project technology performance level (CWT and CWS): For each individual CWT or CWS, it shall be demonstrated at the start of each crediting period with water quality testing reports that the water directly supplied by the project water technology/source	The project activity has undertaken water quality testing for HWT, IWT and community filters to assess the microbial quality of the filtered water. The testing for chemical contaminants for CWT will be undertaken prior to first verification to ensure compliance with applicable

¹² [Water Quality & Effluent Monitoring Guidelines.pdf \(wasreb.go.ke\)](https://wasreb.go.ke/Water_Quality_&_Effluent_Monitoring_Guidelines.pdf)

achieves both: i. microbial quality in line with either (i) national standards or guidelines for microbial quality of drinking water, or in the absence of such requirements, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality (Table 7.10, WHO, 2017); and ii. compliance with (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical contamination and physical and aesthetic aspects.	national standards on priority chemical contamination and physical and aesthetic aspects. This has been approved by the GS through a deviation approval dated 20/12/2021.
The project must conduct annual water hygiene education campaigns for the end-users.	The project activity will organise relevant water hygiene campaigns annually to sensitize end users. The criteria set in the methodology for this purpose shall be followed.
A project applying this methodology may make SDG claims if relevant monitoring parameter(s) is included in the monitoring plan to demonstrate and confirm the project's contributions to SDGs.	The project activity intends to claim contribution towards SDGs 3,6 and 8. The relevant monitoring parameters have been included in the monitoring plan.

B.3. Project boundary

The project boundary is specified in section 3.1 of the applied GS methodology as follows:

The project boundary includes:

- a. the physical, geographical sites of the low- or zero-greenhouse gas-emitting technologies to treat/supply safe drinking water installed by the project activity,
- b. any back-up engines or other equipment using fossil-fuel related to the low greenhouse gas emitting technologies,
- c. the electricity grid, in the case electricity is used by the project, and
- d. the household, commercial and institutional buildings where the end users of safe water provided by the project are located.

The project will introduce zero-energy water filtration technology which will displace the use of woody fuels traditionally used to treat drinking water. The water filtration technologies require no energy input or consumables.

The project boundary is therefore defined as the location of the water filters and the environs within which purified water from the project technologies are consumed. The target area for this project is defined by the regions in which the project activity is to be implemented, which is 32 counties of the Republic of Kenya identified as the project boundary. Refer to section A.2 of the PDD for more details. The greenhouse gases included in or excluded from the project boundary are shown in Table below.

Source		GHGs	Included?	Justification/Explanation
Baseline	Emissions from wood fuels utilized for obtaining safe drinking water displaced due to project activity	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Minor source of emissions
		N ₂ O	Yes	Minor source of emissions
Project scenario	Emissions from wood fuels utilized for obtaining safe drinking water in project scenario	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Minor source of emissions
		N ₂ O	No	Minor source of emissions

B.4. Establishment and description of baseline scenario

Section 3.5 of the applicable methodology (Emission reductions from Safe Drinking Water Supply v.1.0) describes the application of the methodology to water treatment project activities. The project activity qualifies as a Sectoral scope 03, Type III project activity.

The baseline scenario for this project activity is identified as provided under section 3.5 of the applicable methodology. For users that boil unsafe water for drinking in the pre-project scenario, the general baseline scenario is that users would have boiled water for drinking in the absence of the project activity using non-renewable biomass (firewood and charcoal).

Non-renewable biomass has been used in the project region since 31 December 1989. The biomass used in the baseline scenario qualifies under non-renewable definition since most of material is obtained from forests that remain to be forest land, woody and non-woody biomass from crop lands and/or grasslands, biomass residues and sometimes from non-fossil fraction of industrial or municipal waste. To quantify the parameter fNRB the PP has selected to use a conservative approach by calculating the value, where a value of 91.14% has been applied instead of using CDM Default Values. The CDM NRB default value of 92%¹³ was accepted by the CDM EB on 19 September 2012 and was applied as the default value till 18 September 2017 when it expired.

The methodology also provides that for household end-users currently drinking unsafe water, the principles of suppressed demand are applied, such that the general baseline scenario is assumed to be that users would have boiled water for drinking in the absence of the project activity. However, the suppressed demand baseline does not apply to a large-scale project. A large-scale project can only account for the users that boil water in the pre-project scenario. The suppressed demand baseline may be applied for institutional end-users, except where the institution is connected to a public distribution network (PDN) that supplies safe drinking water - unless justified that supplied water quality doesn't meet safe water definition.

¹³ [CDM: Fraction of non-renewable biomass \(fNRB\) \(unfccc.int\)](http://unfccc.int)

Based on the above and given that the project is large scale, the principle of suppressed demand shall not be applied.

1. Pre-project boiling of water.

Access to clean drinking water is a major problem in many parts of Africa including Kenya. Boiling is the most commonly used household water treatment (HWT) method globally, with an estimated 1.2 billion users (~70% of all HWT users¹⁴). The Kenya Demographic and Health Survey Data collected in 2014 shows that 25.5% of urban households and 22.5%¹⁵ of rural households' boil water prior to drinking (see table 2.1 on page 12). The report further reveals that at least 54.5% and 54.1% of urban and rural households do not treat water and hence end up consuming raw water.

Total	100.0	100.0	100.0	100.0	100.0	100.0
Water treatment prior to drinking¹						
Boiled	25.5	22.5	23.7	25.9	21.0	22.7
Bleach/chlorine added	21.7	22.5	22.2	24.0	23.8	23.8
Strained through cloth	0.4	1.1	0.8	0.6	1.3	1.0
Ceramic, sand or other filter	1.2	3.5	2.6	1.6	3.9	3.2
Solar disinfection	0.0	0.0	0.0	0.0	0.0	0.0
Other ²	0.7	2.0	1.5	0.9	2.0	1.6
No treatment	54.5	54.1	54.3	51.9	54.1	53.3
Percentage using an appropriate treatment method³						
	44.9	44.2	44.5	47.5	44.1	45.3
Number of all households	15,290	21,140	36,430	48,946	93,762	142,708
Person who usually collects drinking water						
Adult female 15+	27.7	56.8	44.6	34.3	64.2	53.9
Adult male 15+	16.4	11.8	13.7	10.9	7.7	8.8
Female child under age 15	0.8	2.5	1.8	1.3	2.7	2.2
Male child under age 15	0.6	1.2	1.0	0.8	1.3	1.1
Other	0.8	0.6	0.8	0.8	0.6	0.7
Water on premises	53.5	26.7	38.0	51.8	23.3	33.2
Total	100.0	100.0	100.0	100.0	100.0	100.0
Number of households selected for full questionnaire	7,280	10,080	17,360	23,176	44,073	67,249

Other methods of water treatment identified include chlorination, straining through cloth, and solar disinfection have been tested but with very little success because of the need for maintenance of equipment and continuous purchase of disinfectants.

Of all the methods of water treatment available for both urban and rural households in Kenya, boiling water without any additional treatment method is the most common practice. This data is in line with the baseline survey findings conducted for the

¹⁴ <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5817760/>
¹⁵ <https://dhsprogram.com/pubs/pdf/FR308/FR308.pdf>

project where out of 78/198 (39.4%) households who were treating water, all of them (100%) indicated they were boiling drinking water. The general baseline scenario for households is therefore assumed that users would have boiled water for drinking in the absence of the project activity.

Table 2: Households baseline water treatment practices analysis

Baseline water treatment	Outcome	%
Households treating water	78	39.4%
Households drinking raw water	120	60.6%

Source: Households Baseline surveys tab baseline water treatment

Table 3: Households baseline water treatment methods analysis

Water treatment methods	Outcome	%
Boiling	78	100%
Treatment chemicals	0	0%
Other methods	0	0%

Source: Households Baseline surveys tab baseline water treatment

In Sub-Saharan Africa (SSA), more than 90% of the population relies on wood fuel (i.e., firewood and charcoal) as a primary source of domestic energy. Over 80% of households in urban areas use charcoal, while firewood is mainly used in rural areas and by institutions such as schools. Study shows that Kenya's urban centres are highly reliant on charcoal to meet the population's energy needs. 86 percent of households in Nairobi County use charcoal for cooking and boiling water and 43 percent use it as their main energy source¹⁶. The targeted communities within the project boundary are all located near woodlands and or forests which provide a ready

¹⁶ <https://forestsnews.cifor.org/68884/beyond-bans-toward-sustainable-charcoal-production-in-kenya?fnl=en>

supply of wood fuel. From the baseline surveys conducted, 180/198 (90.9%) and 18/198 (9.1%) of the households surveyed were using firewood and charcoal fuel respectively, while 47/57 (82.5%) of the institutions used firewood and 10/57 (17.5%) used charcoal for cooking and boiling water.

Table 4: Households baseline fuel consumption analysis

Baseline fuels	Outcome	%
Firewood	180	90.9%
Charcoal	18	9.1%

Source: Households Baseline surveys tab baseline fuel consumption

Table 5: Institutions baseline fuel consumption analysis

Baseline fuels	Outcome	%
Firewood	47	82.5%
Charcoal	10	17.5%

Source: Institutional Baseline surveys tab baseline fuel consumption

1. Raw water and suppressed Demand

Section 3.4.2 of the methodology stipulates that “The suppressed demand baseline does not apply to a large-scale project. A large-scale project can only account the users that boil water in the pre-project scenario. The suppressed demand baseline may be applied for institutional end-users, except where the institution is connected to a public distribution network (PDN) that supplies safe drinking water - unless justified that supplied water quality doesn’t meet safe water definition”.

The project activity is largescale and hence will only account for end-users that boil water. From the baseline surveys conducted on institutions, 25/57 (43.9%) institutions were treating water through boiling, while 32/57 (56.1%) were consuming raw water. This is shown in the table below.

Table 6: Institutional baseline water treatment analysis

Baseline water treatment	Outcome	%
Institutions treating water	25	43.9%
Institutions drinking raw water	32	56.1%

Source: Institutional Baseline surveys tab baseline water treatment

Table 7: Institutional baseline water treatment analysis

Water treatment methods	Outcome	%
Boiling	25	100%
Treatment chemicals	0	0%
Other methods	0	0%

Source: Institutional Baseline surveys tab baseline water treatment

Out of the 32 households that were consuming raw water, 20 (62.5%) and 12 (37.5%) of the institutions indicated they would have boiled water and used treatment chemicals respectively.

Table 8: Institutional baseline water treatment analysis

Analysis	Outcome	%
Institutions that would boil water	20	62.5%
Institutions that would use treatment chemicals water	12	37.5%
Institutions that would use other treatment options	0	0.0%

Source: Institutional Baseline surveys_tab baseline water treatment

In developing countries such as Kenya, most institutions do not have a satisfactory level of service in terms of treated water available for consumption due to an inadequate supply of energy to meet their basic water treatment needs. The result of which is less than the required amount of water being treated and subsequently consumed. During the baseline surveys, the institutions were asked to state some of the water, sanitation, and hygiene challenges they face and inadequate access to safe drinking water was the most predominant challenge at 100%. Other challenges

included high incidence of waterborne diseases at 93%; Lack of access to information on water and sanitation hygiene improvement at 89.5%; and Inadequate sanitation facilities at 78.9%.

Table 9: Institutional baseline water, sanitation, and hygiene challenges analysis

WASH Challenges	Yes	No
Inadequate access to safe drinking water	100%	0.0%
Inadequate sanitation facilities	78.9%	21.1%
High incidence of waterborne related illnesses	93.0%	7.0%
Lack of access to information on water and sanitation hygiene improvement	89.5%	10.5%

Source: Institutional Baseline surveys tab basic information

The baseline is thus measured in the project activity scenario i.e., after the introduction of the water treatment technology, as the sum of the amount of raw water treated and the amount of raw water boiled. This represents the amount of treated water per person in institutions that would be consumed after removing the barriers raised by the requirement to boil. This level however is capped at the level WHO states meets the basic needs for treated water. This cap is set at 5.5 litres of water per person per day.

The applied methodology provides dedicated equations to calculate emissions reductions realized from the implementation of a safe water supply project based on the different technology types. This has been applied in estimation of the emission reductions as indicated in the Ex-ante emission reduction estimation spreadsheet.

Selection and justification of the baseline scenario

Section 3.5.1 of the methodology provides that each Project or VPA shall determine the applicable baseline scenario for fuel, technology, and end-user group as applicable.

The project proponent conducted baseline surveys among the target population to identify and document the following pre-project conditions that define the specific baseline scenario of the end-user group(s) of the project:

- a. Pre-project practices of boiling water or drinking unsafe water (suppressed demand): Document the drinking water sources and/or treatment technologies available and used in the project boundary.

Households

The baseline drinking water sources used by the project were identified during the baseline surveys conducted for 198 households. The criteria for the definition of Water Sources provided under annex 2 of the applicable methodology was followed in identifying water sources as either protected or unprotected. The majority of the households were found to be sourcing drinking water from unprotected sources such as surface water (43.9%), unprotected springs (27.3%), and Unprotected wells (21.7%).

Nine households were found to be using piped water into the compound/yard. However, it was observed that the water is pumped directly from surface water sources (rivers), without undergoing any form of treatment and was hence classified as unprotected water sources. Two of the households indicated they boil drinking water, while six indicated they would boil water if prevailing treatment barriers were removed. 1 household would prefer using treatment chemicals. This further demonstrates that the water source cannot be regarded as a protected water source. Three households indicated they harvest rainwater for drinking, though their main source of water is unprotected wells and they only harvest drinking water during the rainy season. The water harvested is never enough to last the households until the next rainy season and hence most of the time consumes unsafe water from other baseline water sources. Two of the households indicated they would boil their water if the prevailing barrier was removed, while one indicated they boil water for drinking

which justifies that the rainwater is not a reliable source of protected water and has therefore been considered as an unprotected water source.

An analysis of the baseline water sources is provided below.

Table 10: Baseline drinking water sources for households

Drinking water sources	Outcomes	Percentage	Comments
Unprotected spring	54	27.3%	Unprotected water source
Surface water (rivers, reservoirs, lakes, ponds, streams, canals, dams, and irrigation channels)	87	43.9%	Unprotected water source
Unprotected well	43	21.7%	Unprotected water source
Water Kiosks	2	1.0%	Protected water source
Rainwater collection	3	1.5%	All the 3 households' main source of drinking water is unprotected wells and they only harvest drinking water during the rainy season. The water harvested is never enough to last the households until the next rainy season and hence most of the time consume unsafe water from other baseline water sources. Classified as an unprotected water source.
Piped water into compound, yard, or plot	9	4.5%	Unprotected water source as the water is pumped directly from the river without any form of treatment

Source: Households Baseline surveys tab baseline water sources

Overall, only 1% of the drinking water sources used by households in the project boundary were found to meet the criteria of protected water sources while 99% were found to be unprotected as analyzed below.

Table 11: Summary of protected and unprotected water sources used by households

Water sources classification	Outcome	%
Protected water sources	2	1.0%
Unprotected water sources	196	99.0%

Source: Households Baseline surveys tab baseline water sources

The distance covered by households to the baseline water sources was also assessed during the baseline surveys. It was found that majority the of the households (54.5%) travel less than 1km to the water source, and only 16.7% travel more than 1km. This is as analyzed in the table below.

Table 12: Distance covered by households to baseline water sources

Distance to water sources	Outcome	%
Less than 1km	108	54.5%
1km	57	28.8%
More than 1km	33	16.7%

Source: Households Baseline surveys tab baseline water sources

As presented in table 2 above, 39.4% of the households surveyed were treating water, while 60.6% were consuming raw water. All the households treating water were using boiling as the treatment method.

Institutions

The baseline water sources for the institutions were also assessed during the baseline surveys. Out of the 57 institutions, 27 (47.4%) had their main source of drinking water as surface water bodies followed by unprotected springs and unprotected wells at 24.6% and 14% respectively. 1 institution (1.8%) had a protected well as the source of drinking water. 1 institution (1.8%) sourced water from piped water into compound, yard, or plot. However, it was established that the source of the piped water is a surface water body (river), and the water is piped to the institution without any form of treatment. Further, the institution stated that they would boil water if the prevailing treatment barrier was removed. The water source has therefore been considered an unprotected water source.

There were also 6 institutions that indicated they harvest rainwater for drinking. However, this is short-term and is only done during the rainy season. It was established that the primary source of drinking water for these institutions are unprotected springs and rivers which are unprotected water sources. Additionally, 3 of these institutions boil their water while 2 indicated they would boil if the prevailing barriers were removed, and 1 would use treatment chemicals. These have therefore

been categorised as unprotected water sources. The summary is presented in table below.

Table 13: Institutions baseline water sources analysis

Drinking water sources	Outcomes	Percentage	Comments
Unprotected spring	14	24.6%	Unprotected water source
Surface water (rivers, reservoirs, lakes, ponds, streams, canals, dams, and irrigation channels)	27	47.4%	Unprotected water source
Unprotected well	8	14.0%	Unprotected water source
Protected well	1	1.8%	Protected water source
Rainwater collection	6	10.5%	All the 6 institutions main sources of drinking water are unprotected springs and river and they only harvest drinking water during the rainy season. Classified as unprotected water source.
Piped water into compound, yard, or plot	1	1.8%	Unprotected water source as the water is pumped directly from the river without any form of treatment.

Source: Institutions Baseline surveys tab baseline water sources

Overall, only 1.8% of the baseline water sources used by institutions surveyed were found to be protected while the rest 98.2% were unprotected water sources.

Table 14: Summary of institutional protected and unprotected water sources

Water sources classification	Outcome	%
Protected water sources	1	1.8%
Unprotected water sources	56	98.2%

Source: Institutions Baseline surveys tab baseline water sources

The distance covered by institutions to the baseline water sources was also assessed during the baseline surveys. It was found that majority of the institutions (77.2%) travel less than 1km to the water source, and only 7% travel more than 1km. This is as analysed in the table below.

Table 15: Distance covered by the institutions to the baseline water sources.

Distance to water sources	Outcome	%
Less than 1km	44	77.2%
1km	9	15.8%
More than 1km	4	7.0%

Source: Institutions Baseline surveys tab baseline water sources

As presented in table 6 above, 43.9% of the institutions surveyed were treating water, while 56.1% were consuming raw water. All the institutions (100%) treating water were using boiling as the treatment method. 62.5% of the institutions indicated they would boil drinking water, while 37.5% stated they would use treatment chemicals if prevailing barriers were removed.

- b. Efficiency of water boiling systems: Document the baseline stove or water boiling technologies and technologies' thermal efficiency used in the project boundary.

Households

The baseline stove technologies used by the households were identified during the baseline surveys. 60.6% of the households used traditional 3 stone firewood stove, while 30.3% used ~~improved-conventional~~ firewood stoves. ~~However, none of the improved stoves was found to have a grate or chimney.~~ 6.6% of the households were using the traditional charcoal stoves, with only 2.5% using improved stoves. On the stove efficiencies, PP has applied the default values provided in the applicable methodology (Parameter SDWS11) for the different baseline stove technologies identified from the surveys as summarised below.

Table 16: Households baseline stove technologies and thermal efficiencies

Baseline stoves	Outcome	Percentage	Thermal efficiency	Comments
Traditional 3 stone firewood stove	120	60.6%	10%	methodology default
Improved Conventional firewood stove	60	30.3%	10 20%	methodology default. None had grate or chimney
Non improved charcoal stove	13	6.6%	20%	methodology default
Improved charcoal stove	5	2.5%	30%	methodology default

Source: Households baseline survey-tab baseline fuel consumption

Sample baseline stove technologies photographs



Figure 7: Dinah Kerubo's traditional 3 stone firewood stove, Nyamira County

Source: Baseline surveys 2021



Figure 8: Lilian Nterena's traditional 3 stone firewood stove, Laikipia County

Source: Baseline surveys 2021



Figure 9: Peres Atieno's traditional 3 stone firewood stove, Kisumu County
Source: Baseline surveys 2021



Figure 10: Godfrey Ogendi's ~~improved conventional~~ firewood stove, Nyamira County
Source: Baseline surveys 2021

Institutions

Majority of the institutions surveyed (45.6%) were using ~~improved-conventional~~ firewood stoves, followed by traditional 3 stone firewood stoves at 36.8%. ~~None of the firewood stoves was found to have a grate or chimney.~~ 12.3% and 5.3% were using non improved and improved charcoal stoves respectively. The PP has applied the default values provided in the applicable methodology (Parameter SDWS11) for the different baseline stove technologies identified from the surveys as summarised below.

Table 17: Institutions baseline stove technologies and thermal efficiencies

Baseline stoves	Outcom e	Percentag e	Efficienc y	Comments
Traditional 3 stone firewood stove	21	36.8%	10%	methodology default
Improved Conventional firewood stove	26	45.6%	10 20%	methodology default. None had a grate or a chimney
Non-improved charcoal stove	7	12.3%	20%	methodology default
Improved charcoal stove	3	5.3%	30%	methodology default

Source: Institutions baseline survey-tab baseline fuel consumption

Sample photograph



Figure 11: Imara Siongirio Academy ~~improved-conventional~~ firewood stove, Bomet County
Source: Baseline surveys 2021

- c. Baseline fuels: Document the baseline cooking fuels used and/or fuels used for water boiling in the project boundary.

Households

The households were found to be using two main fuels including firewood (90.9%) and charcoal (9.1%) for cooking and boiling water.

Table 18: Households baseline fuels

Baseline fuels	Outcome	%
Firewood	180	90.9%
Charcoal	18	9.1%

Source: Households baseline survey-tab baseline fuel consumption

68.7% of the households indicated they collect the baseline fuels from nearby forests or woodlands, while 31.3% were purchasing from vendors.

Table 19: Households' source of baseline fuels

Source of fuel	Outcome	%
Collection	136	68.7%
Purchasing	62	31.3%

Source: Households baseline survey-tab baseline fuel consumption

For the time spent in getting the fuel, most of the households (45.5%) spend 30 to 60 minutes in getting their baseline fuel. 34.3% indicated they spend less than 30 minutes while 20.2% spent more than 1 hour.

Table 20: Time spent by households to get baseline fuel

Time spent getting fuels	Outcome	%
Less than 30 minutes	68	34.3%
30 minutes-1 hour	90	45.5%
More than 1 hour	40	20.2%

Source: Households baseline survey-tab baseline fuel consumption

The PP also assessed variation of fuel consumption during different seasons. During the dry season, 60.1% of the households consumed less fuel, while 87.4% indicated they consume more during the rainy season. During holidays and weekends, 76.8% and 57.6% of the households respectively indicated they consume more fuel. The detailed analysis is provided in the table below.

Table 21: Seasonal variation in baseline fuel consumption among households

Seasonal variation in fuel consumption	Outcome	Percentage
Dry season	Outcome	%
More	43	21.7%
Less	119	60.1%
No change	36	18.2%
Rainy season		
More	173	87.4%
Less	14	7.1%
No change	11	5.6%
Holidays		
More	152	76.8%
Less	24	12.1%
No change	22	11.1%
Weekends		
More	114	57.6%
Less	56	28.3%
No change	28	14.1%

Source: Households baseline survey-tab baseline fuel consumption

Institutions

As the case with households, majority of the institutions (82.5%) were using firewood as their baseline fuel compared to 17.5% who were using charcoal for cooking and boiling drinking water.

Table 22: Institutions baseline fuels

Baseline fuels	Outcome	%
Firewood	47	82.5%
Charcoal	10	17.5%

Source: Institutions baseline survey-tab baseline fuel consumption

The main source of the baseline fuel for the institutions was purchasing from vendors (71.9%), while 28.1% collect from forests.

Table 23: Institution's source of baseline fuel

Source of fuel	Outcome	%
Collection	16	28.1%
Purchasing	41	71.9%

Source: Institutions baseline survey-tab baseline fuel consumption

In terms of time spent in getting the fuel, 45.6% spend less than 30 minutes while 38.6% spend between 30-60 minutes. Only 15.8% spent more than 1 hour in getting fuel.

Table 24: Time spent by institutions in getting the baseline fuel

Time spent getting fuels	Outcome	%
Less than 30 minutes	26	45.6%
30 minutes-1 hour	22	38.6%
More than 1 hour	9	15.8%

Source: Institutions baseline survey-tab baseline fuel consumption

On seasonal variation of the baseline fuel consumption, 52.6% spent less during dry seasons while 87.7% spent more during the rainy season. During holidays and weekends 71.9% and 63.2% respective indicated they spend less fuel.

Table 25: Seasonal variation in fuel consumption by institutions

Seasonal variation in fuel consumption	Outcome	Percentage
Dry season		
More	19	33.3%
Less	30	52.6%
No change	8	14.0%
Rainy season		
More	50	87.7%
Less	3	5.3%
No change	4	7.0%
Holidays		
More	10	17.5%

Less	41	71.9%
No change	6	10.5%
Weekends		
More	11	19.3%
Less	36	63.2%
No change	10	17.5%

Source: Institutions baseline survey-tab baseline fuel consumption

Baseline Survey

Pursuant to Section 4.2 of the applicable methodology, the project developer carried out baseline surveys. The aim of the surveys was to collect data and quantify parameters of interest used to calculate baseline emissions and accurately determine the baseline situation. The baseline surveys were conducted for each baseline scenario including households and institutions' end users. The surveys for the households were conducted between 23/08/2021 to 01/09/2021, while that for institutions were carried out between 23/08/2021 to 10/09/2021. The surveys were conducted by a trained team of data enumerators under the supervision of the Aqua Clara Monitoring and Evaluation officer, Ms. Josephine Orare. Prior to undertaking the surveys, the PP (ClimateCare), had organized a trained with the Aqua Clara monitoring officers, Ms. Josephine Orare and Ms. Risper Mose to go over the baseline survey requirements. The training was held on 18/08/2021 via Microsoft teams and was facilitated by Ms. Lilian Kagume from ClimateCare.

The Aqua Clara team then conducted a physical 1-day training for the 9 data enumerators who were engaged in the exercise. The nine included 6 females and 3 males and were drawn from the different counties where the baseline surveys were conducted. The training was held on 20/08/2021 at the Aqua Clara Kenya office in Rigoma, Nyamira County. The purpose of the training was to go through the baseline survey data requirements and procedure, remind the data collectors of basic data collection skills and qualities of a good data collector, and train the enumerators how to use the M-water app for data collection. A list of the training participants list has been submitted as supporting evidence.



Figure 12: The data enumerators and the ACK M&E officers during a group photo after the training

The M-water app¹⁷ was used for data collection. This is a software app which enables the enumerators to send data from the field in real time. The app is installed in a smartphone and is easy to use making data collection more efficient and less bulky as compared to using printed questionnaires. The PP prepared the template to be used for the surveys, which was then fed on the app by the Aqua Clara team. During data collection, the app automatically sends the data being collected by the enumerators to the Aqua Clara M&E officer for verification and back up. After the data collection exercise, the baseline surveys data was submitted to ClimateCare in an excel format for analysis. A summary of the baseline data collected has been submitted in excel format as a support document.

¹⁷ [mWater](#)



Figure 13: Participants practising how to use the mWater app during the training

Baseline Survey Representativeness

The baseline survey requires in person interviews with a robust sample of end users without project technologies that are representative of end users targeted in the project activity. The baseline surveys were conducted within the applicable geographical areas covered under the project boundary. The surveys were conducted separately for both households and institutions, to ensure the data collected was representative of the end-users. The end-users surveyed were randomly selected and it was established that they were not using the project technologies.

The baseline surveys were conducted in counties where the project had distributed the water filters. Out of the 32 counties that form the project boundary, by the time of the surveys the project had distributed filters in households within 17 counties and institutions spread across 10 counties.

The counties were first put according to their respective provincial boundaries, characteristic by similar social-economic and environmental conditions. The PP then analysed the total number of filters for each technology distributed in each county. The total for household filters (includes BSF and HHFF) and institutions (IHFF and CF) was determined. The sample size per county was then calculated as a proportion of the number of the filters sold for both households and institutions. There were a few counties which had only 1 selected sample and this was added to an adjacent county,

or a county assumed to have similar baseline conditions. This is due to logistics challenge of travelling to the counties just to undertake 1 survey especially in the light of Covid 19 restrictions on movement.

A detailed procedure on the sample selection is provided in the Aqua Clara Baseline Sampling spreadsheet provided as supporting evidence. A summary of the selected samples and the actual surveys conducted per County is provided below.

Table 26: Baseline survey sampling and surveys conducted per county

Samples by County				
County	HH samples	Actual Surveys Conducted	IH samples	Actual Surveys Conducted
Uasin Gishu	3	7	2	2
Narok	10	6	1	2
Laikipia	41	42	7	7
Kericho	2	5	0	0
Bomet	27	30	8	15
Nandi	2	9	0	0
Kisumu	4	17	0	0
Migori	5	12	2	2
Homabay	4	20	2	6
Nyamira	27	36	4	10
Kisii	7	14	7	10
Meru	0	0	1	3
Total	132	198	34	57

Source: Aqua Clara baseline sampling-tab samples per county

In person interviews were held with the end-users to collect the required baseline data. As demonstrated in the table above, the required samples were met in all counties except Narok County where it was impossible to complete the 10 surveys after the data enumerator fell sick.

Baseline Survey Sample Sizing

The methodological requirement is that baseline surveys shall be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

Group size <300: Minimum sample size 30 or population size, whichever is smaller.

Group size 300 to 1000: Minimum sample size 10% of group size

Group size > 1000 Minimum sample size 100

The project has two baseline scenarios, for households and institutions. PP followed the sampling requirements outlined. The group size was taken as the total number of filters installed so far. The group size for households was 1,925 and hence the minimum sample required was 100. A sample of 132 households was selected. Likewise, the group size for institutions was 120 and hence the minimum sample was 30. A sample of 34 institutions was selected.

A summary of the sample selection by county and the actual number of surveys is provided in table 29. It can be noted that in the two scenarios, the actual number of surveys conducted is more than the samples. The extra surveys were done to cater for any attritions in responses during the surveys. For households, 198 surveys were completed and for institutions the actual surveys were 57.

Data Collected

The data collected was specific to the characteristics of each baseline scenario i.e households and institutions, and was tailored to gather information about each of the following:

1. User follow-up

Requirement	Applicability to the Baseline Survey
Address or location	The location of users was established. Household location data was specified at the interview stage and recorded in the survey data in electronic spread sheets. GPS coordinates were recorded for most end-users where network connectivity allowed. Refer to the households and institutions baseline survey spreadsheets tab basic information.
Mobile telephone number and/or landline telephone number (when possible)	Households' telephone numbers (where available) were recorded during the interviews. Refer to the households and institutions baseline survey spreadsheets tab basic information.

2. End user characteristics

Requirement	Applicability to the Baseline Survey
Number of people served by baseline technology	Survey questionnaires included a specific question to assess the number of people in the households, and institutions. Refer to the households and institutions baseline survey spreadsheets tab basic information.
Source of livelihood for the households	Identify the source of livelihood for households. Refer to the households' baseline survey spreadsheets tab basic information for detailed analysis. .
Water, Sanitation and Hygiene challenges in institutions	Survey questions to establish the common WASH challenges in institutions. Refer to the institutions baseline survey spreadsheets tab basic information for detailed analysis.

3. Baseline technology and fuels

Requirement	Applicability to the Baseline Survey
Types of baseline stove technologies used and their thermal efficiency	Dedicated questions were included to assess baseline stove technologies and determine their thermal efficiency. <i>Refer to the detailed summary on table 16 and 17.</i>
Types of baseline fuels used, and estimated proportions used	Survey questionnaires included specific question to know the type of baseline fuels used and the estimated quantity consumed per week. <i>Refer to the detailed summary on table 18 and 22.</i>
Seasonal variations in baseline technology and fuel use	Seasonal variations in fuel consumption patterns varying through rainy, dry seasons, and holidays and weekends were assessed. <i>Refer to the detailed summary on table 21 and 25.</i>
Sources of fuels; (purchased or hand-collected, etc.) and prices paid, or effort made (e.g., walking distances, persons collecting, opportunity cost)	Sources of fuel were assessed with binary options of fuel collection or purchase. Efforts were made to assess with dedicated questions the average distance travelled to collect or purchase baseline fuel. <i>Refer to the detailed summary on table 19 and 23 for sources of fuel, and table 20 and 24 on the time effort.</i>

4. Baseline water sources

Requirement	Applicability to the Baseline Survey
Baseline water sources	Identification of the baseline water sources used for drinking water and categorization on whether they are improved, or non-improved sources was also assessed as part of the survey. <i>Refer to the detailed summary on table 10 and 13.</i>

5. Sustainable Development assessment

Requirement	Applicability to the Baseline Survey
Level of smoke in the households	Dedicated question to assess baseline smoke levels. <i>Refer to the detailed summary on table 27 and 31 below.</i>
Incidence of water borne diseases and Indoor air pollution related incidences	Survey questionnaires to include a specific question to know whether the household has suffered from illnesses related to water contamination and indoor air pollution in the last 30 days. <i>Refer to the detailed summary on table 28 and 32 below for air quality related illnesses, and table 29 and 33 on waterborne illnesses.</i>
Participation in water, hygiene and sanitation improvement trainings, workshops, or seminars	Dedicated question to assess involvement of households and institutions in water and sanitation projects. <i>Refer to the detailed summary on table 30 and 34 below.</i>

Baseline sustainable development parameters

Households

The households' end-users were asked to rate the level of indoor smoke within their households. A majority (60.1%) expressed the smoke level was high while 39.9% rated it as moderate.

Table 27: Level of smoke in the households

Air quality parameters	Outcome	Percentage
Level of smoke		
High	119	60.1%
Moderate	79	39.9%
Low	0	0.0%

Source: Households baseline survey-tab SD assessment

For incidence of indoor air quality related illnesses, coughing and itchy eyes were the predominant illnesses rated at 87.9% and 86.4% respectively.

Table 28: Summary of Households Incidence of indoor air quality related illnesses

Incidence of indoor air quality related illnesses	Yes	No
Coughing	87.9%	12.1%
Itchy eyes	86.4%	13.6%
Eye problems	39.4%	60.6%
Respiratory illnesses	27.8%	72.2%

Source: Households baseline survey-tab SD assessment

Diarrhoea was the most witnessed waterborne related illness encountered by households rated at 97%, followed by cholera (33.8%). This is mostly due to consumption of unsafe water from unprotected water sources from the project boundary.

Table 29: Summary of Households Incidence of waterborne related illnesses

Incidence of waterborne related illnesses	Yes	No
Diarrhoea	97.0%	3.0%
Skin rash	25.3%	74.7%
Cholera	33.8%	66.2%
Foot sores	15.7%	84.3%

Source: Households baseline survey-tab SD assessment

During the surveys, the households were also asked to state whether they had attended any seminar, workshop or training related to water sanitation or hygiene improvement organised by other parties such as other water filtration projects or government agencies. An overwhelming 99% of the respondents had not taken part in such opportunities.

Table 30: Households WASH workshops/seminars participation

Workshop/training/campaign on WASH	Outcome	%
Yes	2	1.0%
No	196	99.0%

Source: Households baseline survey-tab SD assessment

Institutions

82.5% of the institutions surveyed expressed they witnessed high smoke levels while 17.5% rated the smoke level as moderate.

Table 31: Indoor smoke level in institutions

Air quality parameters	Outcome	Percentage
Level of smoke		
High	47	82.5%
Moderate	10	17.5%
Low	0	0.0%

Source: Institutions baseline survey-tab SD assessment

On incidence of indoor air quality related illnesses in institutions, the most common illness was coughing at 94.7%, followed by itchy eyes at 82.5%. Respiratory illnesses were the least experience illness at 26.3%.

Table 32: Summary of Institutions Incidence of indoor air quality related illnesses

Incidence of indoor air quality related illnesses	Yes	No
Coughing	94.7%	5.3%
Itchy eyes	82.5%	17.5%
Eye problems	33.3%	66.7%
Respiratory illnesses	26.3%	73.7%

Source: Institutions baseline survey-tab SD assessment

On waterborne related illnesses, 93% reported diarrhoea as the most common disease followed by cholera at 10.5%.

Table 33: Summary of institutions incidence of waterborne related illnesses

Incidence of waterborne related illnesses	Yes	No
Diarrhoea	93.0%	7.0%
Skin rash	21.1%	78.9%
Cholera	10.5%	89.5%
Foot sores	8.8%	91.2%

Source: Institutions baseline survey-tab SD assessment

During the surveys, the institutions were also asked to state whether they had attended any seminar, workshop or training related to water sanitation or hygiene improvement. 93% of the respondents had not taken part in such opportunities.

Table 34: Institutions WASH workshops/seminars participation

Workshop/training/campaign on WASH	Outcome	%
Yes	4	7.0%
No	53	93.0%

Source: Institutions baseline survey-tab SD assessment

B.5. Demonstration of additionality

Section 3.3. of the applicable methodology for Emission Reductions from Safe Drinking Water Supply v 1.0 provides the following:

"The project developer shall demonstrate that the project could not or would not take place without carbon finance."

The bulk of the project activity's financial requirement is during the development and implementation due to the costs to manufacture and ship the filter membranes for the hollow fibre filters to the host country, to acquire sand for the BSF filters and to meet the filter installation and distribution costs. Over the filters' 5+ year operating life, continued expenditures shall be required to conduct hygiene campaigns with the end-users; conduct the necessary end-user retraining on proper filter usage; and handle maintenance issues and to reinstall the filters properly where necessary.

The project developer is also required to demonstrate additionality by conforming to additionality requirements set under section 3.3.2 of the applicable methodology.

Section 4.1.9 of Community activity service requirements provides that "Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification".

(a) Positive list

(b) Projects located in LDC, SIDS, LLDC4

(c) Microscale projects

Under Annex B on positive list, section 1.1.3 stipulates that *"Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year"*.

The demonstration of additionality has been done following this provision. The project activity can demonstrate that the project technologies have ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year as discussed below.

As per section 3.6.2 of the applied GS methodology, the amount of energy required to boil 1 L of water is 360.83 kJ/L. According to equation 6 and 7 of section B.6 of the PDD, the annual quantity of safe water provided by each of the project water purifier (solely composed of isolated units of the project activity) can be calculated as:

$N_{p,y}$ (Number of premises type p with at least one project technology in year y) * $U_{p,y}$ (Usage rate of the project technology by premises type p during year y) * $Q_{PW_{hh,p,y}}$ (Volume of drinking water per premises p per day in year y) * $DP_{p,y}$ (Days the project technology is present for end-users in the premises p in year y)

For Households

$$Q_y = 1 * 95\% * 28.33 * 365 = 9,823.42L.$$

Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling is ~~11.214.2~~%.

Annual thermal energy savings can be calculated as:

$$360.83kJ/L * 9,823.42L / 1,000,000^{18} / \del{11.214.2}\% = \del{31.6424.96}GJ = \del{26.3520.79}MWh_{th}^{19},$$

which is less than 1,800 MWh_{th} (600MWh).

For institutions

$$Q_y = 1 * 95\% * 600 * 365 = 208,050L.$$

Based on the results of baseline survey, the weighted average efficiency of the baseline stove for water boiling is ~~12.316.8~~%.

Annual thermal energy savings can be calculated as:

$$360.83kJ/L * 208,050L / 1,000,000 / \del{12.316.8}\% = \del{610.33446.84}GJ = \del{508.40372.22}MWh_{th},$$

which is less than 1,800 MWh_{th} (600MWh).

Besides, End-users of the water purifiers are households and institutions. The project activity falls under section 1.1.3 of Annex B Positive list of *GS4GG Community*

¹⁸ 1PJ=10¹⁵J.

¹⁹ Unit conversion from GJ to MWh_{th}=3/3.6=0.833, i.e., 1GJ=0.833MWh_{th}.

Services Activity Requirements (Version 1.2). As per section 4.1.9(a) of *GS4GG Community Services Activity Requirements (Version 1.2)*, the project is considered as deemed additional.

B.5.1 Prior Consideration

According to Gold Standard's Principles and Requirements (version 1.2), Retroactive projects shall submit the required documents for preliminary review (time of first submission) within one year of the project start date. Retroactive Project submitted at a date later than one year from the project start date will not be eligible for Gold Standard certification.

The project start date is 01/05/2020, and PP submitted the required documents for preliminary review on 28/04/2021 which is within one year of the project start date. Therefore, the project is eligible for Gold Standard certification.

The main action made for the project implementation are generalized in the following:

Project milestones

Timeline	Milestone
01/05/2020	Start date of the project (first purchase receipt for the water purifier sold to an end-user).
28/04/2021	The first submission date to GS
21/07/2021	Stakeholder consultation physical meeting
11/08/2021	Project listing by the GS
24/08/2021	Kicked off second round stakeholder feedback round (online)
24/10/2021	Conclusion of stakeholder feedback round
08/09/2021	VVB contracting for validation

B.5.2 Ongoing Financial Need

The project activity is neither under Design Certification Renewal nor required to demonstrate financial additionality. Ongoing financial need as demonstrated below.

The primary goal for the project is the commitment to maximize the provision of clean and safe water over the long term, which requires significant ongoing financial support which will only be attained through carbon financing.

Through the project, more people will be provided with safe drinking water and the carbon financing will support the project through:

- Payment of critical costs such as monitoring and evaluation to ensure consistent and continuous use of water filters thus improving impact in terms of the health outcomes in the community. The finance will also be utilised in organising hygiene campaigns and other WASH related trainings and workshops for communities.
- Payment for staff salaries, making Aqua Clara financially sustainable, ensuring that the organization continues to exist to provide more people with safe drinking water as well as to replace and repair water filters of existing customers.
- Expansion to new areas to increase project coverage to new geographical areas.
- Funds from carbon credits will support research and development towards improving the quality of water filters and cost reduction improvements to reduce the selling price to the end consumer who is mainly a rural dweller and low-income urban dwellers whose average income is less than \$5/day.
- Carbon credits will also enable the project developer to conduct evaluations of the social impact that the usage of the filters is generating. A great reputation with existing and potential partners and clients (especially those inclined to purchase/partner with organizations with eco-friendly products) given that the Aqua Clara filters are certified to reduce carbon emissions.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	Target 13.1: Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries	Total emission reductions
SDG 8	Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.	Number of people employed by the project
SDG 6	Target 6.1: By 2030, achieve universal and equitable access to safe and affordable drinking water for all. Target 6b: Support and strengthen the participation of local communities in improving water and sanitation management	- Number of people attending hygiene education campaigns. - Number of people attending training/ workshops on filter installation and maintenance.
SDG 3	Target 3.3: By 2030, end the epidemics of AIDS, tuberculosis, malaria and neglected tropical diseases and combat hepatitis, water-borne diseases, and other communicable diseases. Target 3.9: By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	- Reduction in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases - Improvement in Indoor Air Quality

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact.

SDG contribution and project monitoring indicator	Project monitoring indicator	Baseline outcome	Project outcome
SDG 3 Good Health and Well-Being	Reduction in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases.	In the baseline scenario, no project water filters were being used. Therefore, the baseline outcome is high incidence of waterborne diseases such as diarrhea (93%), with 0 reduction.	The number of waterborne diseases is expected to decline significantly by up to 100% once the end-users start using the project water filters. Parameter to be monitored during annual project survey. The net benefit of SDG3 = Project outcome of SDG3- Baseline outcome of SDG3
SDG 3 Good Health and Well-Being	Improvement in Indoor Air Quality	In the baseline scenario, no project water filters were being used and household and institutions were using woody	The level of smoke in households and institutions is expected to decline significantly by up to 100% once the end-users start using the project water filters. Parameter to be monitored during annual project survey. The net benefit of SDG3 = Project outcome of SDG3- Baseline outcome of SDG3

		biomass to boil water for drinking. Therefore, the baseline outcome is high smoke levels as reported by 60.1% of households and 82.5% of institutions										
SDG 6 Clean Water and Sanitation	Total number of person.days served with satisfactory level of safe water.	In baseline scenario, no project water filters were being used. Therefore, the baseline outcome benefit is 0.	In the project scenario, total number of person.days served with satisfactory level of safe water can be calculated as follows: Project outcome of SDG 6 = $N_{p,y} * U_{p,y} * QPW_{hh,p,y} * DP_{p,y}$ (6) Where:									
			<table><tr><th>Parameters</th><th>Description</th><th>Source/Value</th></tr><tr><td>Project outcome of SDG 6</td><td>Total number of person.days served with satisfactory level of safe water</td><td>Calculated based on equation (6)</td></tr><tr><td>$N_{p,y}$</td><td>Number of premises type p with at least one project technology in year y</td><td>Sales database</td></tr></table>	Parameters	Description	Source/Value	Project outcome of SDG 6	Total number of person.days served with satisfactory level of safe water	Calculated based on equation (6)	$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database
			Parameters	Description	Source/Value							
			Project outcome of SDG 6	Total number of person.days served with satisfactory level of safe water	Calculated based on equation (6)							
$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database										

			$HN_{p,y}$	Number of individuals per premises type p in year y	Project survey, for ex ante estimation, result of 5.15 from baseline survey has been applied for households.
			$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y	To be monitored, for ex ante estimation, $DP_{p,y}=365$
			$U_{p,y}$	Usage rate of the project technology by premises type p during year y .	To be monitored, for ex ante estimation, $U_{p,y}=95\%$
			$M_{q,y}$	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial	Water quality test, for ex ante estimation, $M_{q,y}=100\%$ based on the manufacturer's specifications of the project technologies

			quality standard for drinking water of Kenya.	
			The net benefit of SDG6 = Project outcome of SDG6 – Baseline outcome of SDG 6.	
SDG 6 Clean Water and Sanitation	Number of people attending hygiene education campaigns.	In the baseline scenario, no project water filters were being used, hence no hygiene campaigns had been organized by the project. Only 7% and 1% of institutions and household reported attending a WASH training or workshop in the baseline. Therefore, the baseline outcome benefit is 1% for household and 7% for	Parameter to be monitored during annual hygiene campaign survey. The net benefit of SDG6 = Project outcome of SDG6 – Baseline outcome of SDG 6.	

		institutions .	
SDG 6 Clean Water and Sanitation	Number of people attending training/ workshops on filter installation and maintenance.	In baseline scenario, no project water filters were being used, hence no training/ workshops on filter installation and maintenance had been organized by the project. Therefore, the baseline outcome benefit is 0.	Parameter to be monitored during annual project survey. The net benefit of SDG 6 = Project outcome of SDG 6 – Baseline outcome of SDG 6.
SDG 8 Decent Work and Economic Growth	Number of people employed by the project	In the baseline situation, no persons had been employed by the project. Therefore, the baseline outcome benefit is 0.	In the project situation, the project shall create jobs for both males and females. The number of employees including their age, gender, and level of remuneration shall be monitored. Net benefit of SDG8=Project outcome of SDG8 – Baseline outcome of SDG8

SDG 13 Climate Action	GHG emission reductions	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0) provides specific surveys, calculations for baseline and project emissions as well as guidance on applicable surveys and tests.
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Specific Calculation Method for Outcome of SDG 13

Baseline emissions

The “Methodology for Emission Reductions from Safe Drinking Water Supply v.1.0” will be applied for this project. The parameters for estimating the SDG impact are as summarized below.

Mq,y Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard requirements specified in relevant microbial quality standard for drinking water of the host country.

The project developer shall conduct water quality testing. In line with the national drinking water standard, all water directly intended for drinking must not have detectable E.Coli in any 100 ml sample i.e., less than 1 Colony Forming Unit (CFU) of E.Coli /100 ml. Field testing kits based on Colony Forming Unit method or Most Probable Number method will be used, in fulfillment with the set requirement for using the method.

$X_{cleanboil,y}$ Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year y

During installation of the filter, the installer sensitizes the end-user on the proper use and maintenance of the filter. The instructions are geared towards reduction of domestic fuel wood consumption and the impracticality of boiling water already treated since it is safe for consumption. This parameter will be monitored during the annual usage survey.

$Q_{m,y}$ Monitored quantity of safe water provided by the CWT project in year y

The CWT units will be fitted with Flow meters which measures water volume directly. The meter readings will be recorded daily, and this will be monitored during the annual usage surveys. Manufacturer guidelines for calibration and maintenance of the measurement devices will be observed.

QPW_p Volume of drinking water per person per day for premises type p

Option 1: allows the application of the default value per person. In the case of institutions, such as schools, the value should reflect the expected drinking water use per person while on the premises of the institution, in line with the following defaults:
- Full-day premises: 4 L/person/day.

- Boarding school: 4 L/person/day
- Half-time premises: 3 L/person/day

Option 2: allows PDs to conduct the Water Consumption Field Tests. In all cases, the value is capped at 5.5 L/person/day.

$HN_{p,y}$ *Number of individuals per premises type p in the project boundary in year y*

The PP shall carry out an annual project survey to determine the number of individuals per premises served by the project technology. The value applied shall be cross-checked against at least one other source listed under the data sources.

$HH_{p,y}$ *Number of premises type p served by the project in year y*

For CWT technology, PP shall conduct annual survey of the premises (e.g., households, schools) within 1 km distance of project water source to check how often the premises used the project water source during the year. This survey may be part of the project survey and may be performed in person, by telephone, by messaging (e.g., text, app), appropriate to the context. Premises that report at least every-two-days use may be counted.

$Do_{p,y}$ *Days the project technology is operational for end-users in premises p in year y*

For CWT units, PP shall demonstrate from log of operation and maintenance system the number of days the technology has been used in the year. For schools and other institutions, as applicable, the days shall also be limited by the number of school days in the period, taking into account weekends and holidays.

$N_{p,y}$ *Accumulated number of premises type p with at least one individual project technology in year y*

The project developer shall keep an updated sales record which will be used to monitor the number of end-users with at least one project technology.

$U_{p,y}$ *Usage rate of the project technology by premises type p during year y*

The project developer shall conduct annual usage survey on the premises with a project technology to determine the usage rate of the project technology during the year. Under Option 1: where In-person survey of project premises (e.g., households, schools) is conducted, households that show at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 100% to get $U_{p,y}$.

Under Option 2: where Surveys are performed by telephone or messaging (e.g., text, app), households that report at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 75% to get $U_{p,y}$.

The usage survey for household water treatment systems shall be conducted in line with the Usage Surveys Guidelines outlined in Annex 1 of the applicable methodology.

$t_{p,y}$ *Usage time of the project technology by premises type p in year y*

PP shall carry out annual surveys to determine the usage time of the project technologies in the case of HWT and IWT units.

$DP_{p,y}$ *Average days the project technology is present for end-users in the premises p in year y .*

PP shall examine the Sales or distribution records annually. Based on the sales or distribution records of "Date of sale/distribution" and ex-ante parameter "Expected technical life of project technology," PP shall determine for each project device how many days of the 365 days of the year it was in the premises and within its technical life. PP shall then calculate the average for all the project technology by premises type p to obtain this parameter.

$DN_{p,y}$ *Average number of individual project technologies in each project premises type p in year y .*

PP shall examine the Sales or distribution records annually. Based on the sales or distribution records of "Quantity of project technologies sold" and identifying information of buyer/recipient, PP shall calculate the average number of project devices per premises. If the project covers different types of end-users (e.g., households, institutions), the average number must be determined per premises type p .

LE_y *Leakage emissions during year y*

The PP has demonstrated in section B.7.3 of the PDD that there is no perceived leakage from the project. However, PP shall monitor the sources of leakage in every two years following the criteria set in the methodology.

Baseline emission factor calculations

As per section 3.6 of the applied GS methodology, baseline emissions can be calculated as follows:

$$BE_y = EF_b * (1 - C_b - X_{cleanboil,y}) * Q_y * M_{q,y} \quad (3)$$

Where:

Parameters	Description	Source/value
BE_y	Baseline emissions from the use of fuel to obtain safe water in the baseline (tCO ₂ e)	
EF_b	Emission factor for the use of fuel to obtain safe water in the baseline (tCO ₂ e/L)	Calculated as following Equation (1)
C_b	Proportion of project households who in the baseline were already using a safe water supply that did not require boiling it (%)	Baseline survey, 2.02% (households) and 3.5% (institutions)
$X_{cleanboil,y}$	Proportion of project households that boil safe water in the project year y (%)	Usage survey, for ex-ante estimation, it assumes a value of 0%. The actual value will be monitored.
Q_y	Quantity of safe drinking water provided by the project in year y (L)	Calculated based on Method 2 for household water treatment technologies.
$M_{q,y}$	Modifier for the water quality in year y	For ex-ante estimation, it assumes a value of 100%. The actual value will be determined based on water quality test.

$$EF_b = SE_{w,b,y} * \sum_f (x_f * (EF_{b,f,CO2} * f_{NRB,f,y} + EF_{b,f,nonCO2})) \div 10^9 \quad (1)$$

As per results of baseline survey, among those who treat drinking water by boiling, the main fuel used are non-renewable biomass of firewood and charcoal EF_b can be calculated as the following equation.

$$EF_b = SE_{w,b,y} * (x_{firewood} * (EF_{b,firewood,CO2} * f_{NRB} + EF_{b,firewood,nonCO2}) + x_{charcoal} * (EF_{b,charcoal,CO2} * f_{NRB} + EF_{b,charcoal,nonCO2})) \div 10^9 \quad (1)$$

Where:

Parameters	Description	Source/value
EF_b	Emission factor for the use of fuel to obtain safe water in the baseline (tCO ₂ e/L)	Calculated as Equation (1), 0.0003 08 (households), 0. 00037 <u>00027</u> (institutions and community filters)
$SE_{w,b,y}$	Specific energy required to boil water (kJ/L)	Calculated as Equation (2), <u>2541.05</u> 3221.69 -(households), <u>2147.79</u> 2933.57 (institutions and community filters)
$x_{firewood}$	Proportion of firewood used in the baseline (fraction)	Baseline survey, 90.9%(households), 82.5% (institutions and community filters)
$x_{charcoal}$	Proportion of charcoal used in the baseline (fraction)	Baseline survey, 9.1%(households), 17.5% (institutions and community filters)
$EF_{b,firewood,CO2}$	CO ₂ emission factor from use of firewood (tCO ₂ /TJ)	IPCC 2006 default value for wood (Table 1.4 of Chapter 1 Introduction), 112 tCO ₂ /TJ
$EF_{b,firewood,nonCO2}$	Non-CO ₂ emission factor arising from use of firewood (tCO ₂ e/TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 9.46t/TJ
$EF_{b,charcoal,CO2}$	CO ₂ emission factor from use of charcoal (tCO ₂ /TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 165.22 tCO ₂ /TJ (includes charcoal production emissions)
$EF_{b,charcoal,nonCO2}$	Non-CO ₂ emission factor arising from use of charcoal (tCO ₂ e/TJ)	<i>Methodology for Emission Reductions from Safe Drinking Water Supply</i> (Version 1.0), 44.83t/TJ (includes production emissions of CH ₄ and N ₂ O)
$f_{NRB,f,y}$	Fractional non-renewability status of woody biomass fuel during	Calculated using CDM TOOL 30, Version3, 91.14% and fixed for the first crediting period.

	year y , in case the baseline fuel is biomass or charcoal	
f	Index for baseline fuel types	Baseline survey, the baseline fuel types are firewood and charcoal

The specific energy required to boil water using the baseline technology ($SE_{w,b,y}$) is determined as follows, by calculating the energy input required to obtain 1 L of boiling water, including boiling and vaporization losses.

$$SE_{w,b,y} = 360.83 / \eta_{wb} \quad (2)$$

Where:

360.83	Default amount of energy required to boil 1 L of water from a first principles approach
η_{wb}	Efficiency of the baseline water boiling (%). Weighted average of baseline stove types. Default efficiency: 10% for three-stone stove, 20% for conventional stove, 30% for improved stove. Based on the results of baseline survey, η_{wb} for households is 11.2 14.2%, and 12.3 16.8% for institutions

Baseline Emissions Calculations

The quantity of safe drinking water provided by the project (Q_y) can be calculated using Method 2, which applies to projects using household and institutional water treatment technologies.

$$Q_y = \sum_p N_{p,y} * U_{p,y} * QPW_{hh,p,y} * DP_{p,y} \quad (6)$$

Where:

Parameters	Description	Source/value
$N_{p,y}$	Number of premises type p with at least one project technology in year y	Sales database
$U_{p,y}$	Usage rate of the project technology by premises type p during year y (%)	Usage rate survey. For ex-ante estimation, $U_{p,y}$ =95% (households and institutions)

$QPW_{hh,p,y}$	Volume of drinking water per premises p per day in year y (L)	Calculated as Equation (7), 28.32(households), 600 (institutions)
$DP_{p,y}$	Days the project technology is present for end-users in the premises p in year y	The actual value will be determined based on the sales database. For ex-ante estimation, it assumes a value of 365.

The volume of drinking water per household per day is determined by considering whether the capacity of the project device is sufficient to provide at least the default amount of drinking water, as follows.

$$QPW_{hh,p,y} = \min((q_i * t_{p,y} * DN_{p,y}), (QPW_p * HN_{p,y})) \quad (7)$$

Where:

Parameters	Description	Source/value
q_i	Capacity of the household water treatment technology (L/h)	Manufacturer specifications, 60
$t_{p,y}$	Usage time of the project technology by premises type p in year y (h/day)	Option 3 Default value of 5 hours applied for ex-ante estimation.
$DN_{p,y}$	Average number of individual project technologies in each project premises type p in year y	The actual value will be determined based on sales database. For ex-ante estimation, it assumes a value of 1 for households, 2 for institutions.
$HN_{p,y}$	Number of individuals per premises type p in year y	The actual value will be determined based on project survey. For ex-ante estimation, result of 5.15 for households and 450 for institutions from baseline survey has been applied.
QPW_p	Volume of drinking water per person per day for premises type p (L)	The actual value will be determined based on Water Consumption Field Tests. For ex ante estimation, the maximum cap of 5.5 L/person/day has been applied.

The quantity of safe drinking water provided by the project (Q_y) can be calculated using Method 1, which applies to projects using community water treatment technologies.

$$Q_y = \min(Q_{m,y}, Q_{pop,y}) \dots \dots \dots (4)$$

Where:

Parameters	Description	Source/value
Qm,y	Monitored quantity of safe water provided by the project in year y (L).	Value of 3,850 applied for ex-ante estimation
$Qpop,y$	Quantity of safe drinking water that could be consumed by project end-users in year y (L)	Calculated as Equation (5), value of 3,850 applied for ex-ante estimation

Quantity of safe drinking water is determined as follows;

$$Q_{pop,y} = \sum_p HH_{p,y} \times HN_{p,y} \times QPW_p \times DO_{p,y} \dots \dots \dots (5)$$

Where:

Parameters	Description	Source/value
$HH_{p,y}$	Number of premises type p served by the project in year y	The actual value will be determined based on sales database. For ex-ante estimation, it assumes a value of 1 since most of the Community filters are installed in institutions.
$HN_{p,y}$	Number of individuals per premises type p (e.g. household, school) in year y	The actual value will be determined based on project survey. For ex-ante estimation, result of 700 from baseline survey has been applied.
QPW_p	Volume of drinking water per person per day for premises type p (L). Apply the default value or monitored value through water consumption field tests in the project scenario, capped at 5.5 L per person per day.	The actual value will be determined based on Water Consumption Field Tests. For ex ante estimation, the maximum cap of 5.5 L/person/day has been applied.
$DO_{p,y}$	Days the project technology is operational for end-users in premises p in year y	The actual value will be determined. For ex-ante estimation, it assumes a value of 347 days is applied which is the maximum number allowed for

		technologies using flow meters for direct measurement.
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Project emissions

As detailed under section 3.7 of the applied GS methodology, project emissions may result from the operation of new low-emission water treatment technologies. The Aqua Clara water filters introduced by the project require no energy input or consumables, which will offer an affordable, long-term and zero emission solution for households and institutions that generally consume unsafe drinking water. No fossil fuel or electricity will be used in the project activity, project emissions (PE_y) is 0 tCO₂e.

Leakage emissions (LE_y)

As provided under section 3.8 of the applied GS methodology, leakage relating to the non-renewable woody biomass shall be assessed and other types of leakage are excluded for simplification. Leakage emissions are investigated in section B.7.3 of the PDD. There is no perceived leakage from the project activity, leakage emissions (LE_y) is 0 tCO₂e.

GHG Emission Reductions

$$ER_y = BE_y - PE_y - LE_y \quad (11)$$

Parameters	Description	Source/value
ER_y	Emission reductions in year y (tCO ₂ e/yr)	Calculated as Equation (11)
BE_y	Baseline emissions in year y (tCO ₂ e/yr)	Calculated as Equation (3) (4), (5), (6), (7),
PE_y	Project emissions in year y (tCO ₂ e/yr)	Calculated as Equation (8) 0 tCO ₂ e
LE_y	Leakage emissions in year y (tCO ₂ e/yr)	0 tCO ₂ e

B.6.2 Data and parameters fixed ex ante

a. Related to water quality.

Data/parameter	SDWS 1: SDG 13 Number of household/institutions per CWT
Unit	-Coordinates of CWT - Acceptable formats for GPS coordinates include DMS (degrees, minutes, and seconds), DMM (degrees and decimal minutes), and DD (decimal degrees) - Number of end-user premises, quantity
Description	End users' premises (e.g., households, institutions) within 1 km distance of project water source or a total collection time of 30 minutes or less for a round trip, including queuing, using the travel modes of walking or pedalling.
Source of data	Sales/Installation database
Value(s) applied	10 CWT units
Choice of data or Measurement methods and procedures	This will be recorded for each CWT installation ex-ante at the time of start of crediting period. In case of progressive installation for new CWT units before 1st issuance for new units.
Purpose of data	Emission reduction calculation
Additional comment	Applies only for CWT where water is retrieved from the CWT location. All end-users included in surveys or sampling related to baseline determination or operation of the project shall be located within the 1 km radius of the CWT location. By the time of this validation, 10 community filters had been installed. All the filters are installed within institutions (schools) and hence the end-users are within a 1km distance to the safe water source provided by the filter. The actual population served by the filters has been provided in the community filters database and the same shall be confirmed prior to the first verification.

Data/parameter	SDWS 2: SDG 13 Project technology description
Unit	NA
Description	The detailed description of the planned project technology
Source of data	HWT and IWT The methodology provides that any of the following sources shall be used: - Manufacturer specifications - Third-party certification by a qualified entity, for example recognised certification agency by National/ International Standard body - Commercial guarantee CWT and CWS: Any of the following sources shall be used: - Manufacturer specifications - Commercial guarantee - Technical reports from the installer - Third-party certification by a qualified entity, for example recognised certification agency by National/ International Standard body The PP has applied the following data sources for the filter technologies: -Manufacturer specifications: sourced from the Aqua Clara filters fact sheets provided as support documents -Third-party certification: KEBS filter test results and certificate provided as support document.
Value(s) applied	i) Technology description HWT and IWT: - Manufacturer name: Aqua Clara International - Product name: Aqua Clara BioSand filters (BSF) and Aqua Clara Hollow fibre filters (for households and institutions) (HHFF and IHFF). - Technology type: Hollow fibre technology (IHFF and HHFF), and Fine sand filtration (for BSF filters) ii) Performance target classification as per the WHO International Scheme to Evaluate

	Household Water Treatment Technologies or proof of compliance with an applicable national standard or guideline for drinking water treatment technologies. This has been demonstrated using the second option <i>"Proof of compliance with an applicable national standard or guideline for drinking water treatment technologies"</i>. The Aqua Clara Water filters have been tested for filtration efficacy for the parameters E. coli, - <i>Staphylococcus Aureas</i> and Salmonella by the Kenya Bureau of Standards (KEBS) and have been found to remove the bacteria by up to 99.9%. This is in compliance with the drinking water standard for the host country which requires that drinking water should not have detectable E. coli. CWT (Community filters): - Manufacturer name: Aqua Clara International - Product name: Aqua Clara Community Filters - Technology type: Hollow fibre membrane - Capacity: Up to 24 L / minute
Choice of data or Measurement methods and procedures	The data applied is based on the manufacturer specifications for the different filter technologies and third-party certification from the national standard for the host country (KEBS).
Purpose of data	Description of project technology
Additional comment	The test results done by the Kenya Bureau of Standards (KEBS) and the certification issued has been provided (See the KEBS Lab test result documents).

Data/parameter	SDWS 3: SDG 6 Project technology performance level (CWT)
Unit	N/A
Description	The water directly supplied by the water source (CWT) must comply with a) Microbial quality in line with: (i) national standards or guideline for microbial quality of drinking water, or in their absence, (ii) the guideline values for verification of microbial quality from the Guidelines for drinking-water quality, 4th edition (Table 7.10, WHO, 2017); and b). Chemical quality (i) national standards or guidelines on priority chemical contamination and physical and aesthetic aspects, or in the absence of such requirements, (ii) international standards or guidelines on priority chemical contamination and physical and aesthetic aspects.
Source of data	GS11169 CF WQT surveys spreadsheet -tab water quality testing.
Value(s) applied	The water supplied by the community filters was tested for microbial quality. The national drinking water standards for Kenya provides that drinking water should have nil E.coli counts/100ml. ²⁰ . Water quality testing was conducted for 8 community filters that had been installed by the time of testing. Out of the 8 samples collected (100%), none was found to have any detectable E. coli. Testing for chemical quality shall be undertaken for community filters prior to the first verification. Initially the PP had applied TPDDTEC v3.1 during the preliminary review stage which did not have a requirement on chemical testing. However, the new methodology came into place when the project was under the review and the PP had to switch to the new meth when the microbial

²⁰ <https://www.waterfund.go.ke/toolkit/Downloads/2.%20Drinking%20Water%20Guidelines.pdf>

Choice of data or Measurement methods and procedures	testing had already been completed. Due to the challenges imposed by the COVID 19 pandemic, it was not possible to go back to the field to collect the water samples hence it was not possible to undertake the chemical testing by the time of validation. PP will however ensure the water quality testing is completed prior to the first verification and the data submitted to the GS. The PP has obtained a deviation approval dated 20/12/2021 from the GS to allow for chemical testing to be done by the time of the first verification. The microbial quality shall be tested using the 3M™ Petrifilm™ E. coli/Coliform Count Plates. The testing shall be done by a team of trained enumerators to ensure the water quality testing procedures are followed. The chemical testing shall be done in laboratories used which must be approved by local health authorities and/or have quality accreditation. The laboratory used shall have evidence to demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures. Table 4.6 Checklist for effective analytical quality assurance of WHO Guidelines, 1997 may be used as a guideline for laboratory compliance with quality assurance practices.
Purpose of data	Determination of Project technology performance level for community filters.
Additional comment	The chemical testing shall be undertaken once at the start of the crediting period, and microbial quality at the CWT location shall be retested following an event that could lead to contamination of the source water (e.g., flooding). On the maintenance side, the Aqua Clara Community water filters use hollow fiber filtration technology and are maintained through a simple process called backwashing. The backwashing of the filter is done every day after use following the procedure outlined in the Aqua Clara CF filter manual provided as supporting evidence. The manual is provided to the recipients of the CF filters, and they are also trained how to do the maintenance during the installation of the filter. The maintenance records will be provided during the monitoring of the project as required.

Data/parameter	SDWS 4: SDG 6
	Regulatory framework for safe water supply
Unit	NA
Description	List and provide a summary of any national, sub-national and local regulations or guidance for safe drinking water supply, operation, and maintenance, including any tariff requirements. Describe how the project complies with the regulatory framework.
Source of data	National, sub-national and local authorities
Value(s) applied	The analysis for regulatory framework for safe water supply has been done and is provided under section A.1 of the PDD. From the analysis, the project has not been found to undermine or conflict with any national, sub-national and local regulations or guidance for safe drinking water supply, operation, and maintenance, including any tariff requirements.
Choice of data or Measurement methods and procedures	The analysis was informed by conducting a detailed review of the existing national and local regulations and guidance of drinking water supply.
Purpose of data	To assess whether the project undermines or conflicts with any national, sub-national and local regulations or guidance for safe drinking water supply, operation, and maintenance, including any tariff requirements.
Additional comment	The regulatory framework shall be updated at the start of each crediting period to demonstrate compliance. This analysis has been done as summarised in the table below. The project is compliant with applicable regulatory framework and shall not undermine or conflict with any national, sub-national, and local regulations or guidance for safe drinking water supply, operation, and maintenance, including any tariff requirements.

Regulatory framework for safe water supply in Kenya

Applicable regulation	Provisions of the law	Project's compliance
Environmental Management and Coordination (Water Quality) Regulations, 2006 ²¹	These Regulations apply to drinking water, water used for industrial purposes, water used for agricultural purposes, water used for recreational purposes, water used for fisheries and wildlife, and water used for any other purposes. Part II of the regulations provides that "All operators and suppliers of treated water, containerized water and all water vendors shall comply with the relevant quality standards in force as promulgated by the relevant lead agencies". Under the first schedule on Quality Standards for Sources of Domestic	The project shall ensure compliance to this act through regular water testing in line with the set water quality standards.

²¹ [WATER QUALITY REGULATIONS \(waterfund.go.ke\)](http://waterfund.go.ke)

	Water, the regulations set out the parameters and maximum allowable limits, key of which is E. coli which should not be detectable.	
The Water Act, 2012 ²²	This is an Act of Parliament to provide for the management, conservation, use and control of water resources and for the acquisition and regulation of rights to use water; to provide for the regulation and management of water supply and sewerage services. The act provides for the establishment of the Water Resources Management Authority (WRMA) and Water Services Regulatory Board (WASREB). The Water Act 2002 under section 47 requires WASREB to	The project is a water filtration project which seeks to promote access to clean and safe drinking water. The project shall not cause any interference with water resources as the water filters are point of use and are distributed for households and institutional use and will not require extraction of water. The project shall also ensure compliance with the set water quality standards as stipulated in the act.

²² [WaterAct_Cap372_.pdf \(kenyalaw.org\)](#)

	determine standards for the provision of water services to consumers and to monitor compliance with established standards for the design, construction, operation, and maintenance of facilities for water services.	
Guidelines on Drinking Water Quality and Effluent Monitoring, 2008 ²³ .	The purpose of this guideline is to: -Promote transparency in the methods of water quality monitoring employed by the Water Service Providers (WSPs). -Ensure through regular monitoring that the quality of water provided meets the standards set by KEBS. -Create awareness among the Water Services Boards (WSBs) and Water Service Providers	The project has obtained the Kenya Bureau of Standards (KEBS) certification and mark of quality. KEBS is government agency for the provision of Standards, Metrology and Conformity Assessment (SMCA) services. The filters have been tested for filtration efficacy for the parameters E. coli, - <i>Staphylococcus Aureas</i> and Salmonella and have been found to remove the bacteria by up to 99.9%. (See attached KEBS Lab test result documents).

²³ [Water Quality & Effluent Monitoring Guidelines.pdf \(wasreb.go.ke\)](https://www.wasreb.go.ke/Water_Quality_&_Effluent_Monitoring_Guidelines.pdf)

	(WSPs) on the water quality monitoring requirements. - Ensure that all WSBs and WSPs follow a systematic way of water quality monitoring -Ensure a minimum standard of water quality monitoring at acceptable costs. Schedule 5 on Microbiological limits for drinking water and containerized drinking water provides that drinking water shall not have detectable E. Coli.	
The Water Act, 2016 ²⁴	The act provides for the regulation, management and development of water resources, water, and sewerage services, and for other connected purposes. Part IV of the act provides that "Every person in Kenya has	The project seeks to accelerate access to clean and safe water by providing efficient, affordable, and reliable water filtration solutions to end-users.

²⁴ [WaterAct_No43of2016.pdf \(kenyalaw.org\)](#)

	the right to clean and safe water in adequate quantities and to reasonable standards of sanitation as stipulated in Article 43 of the Constitution".	
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Data/parameter	SDWS 5: SDG 6
Unit	Water sources in the project boundary
Description	Identify the water sources in the project boundary, and identify whether they are used for drinking water, and for all that are used for drinking water, classify them as improved and unimproved water source.
Source of data	Households baseline surveys and Institutions' baseline surveys spreadsheets tab Baseline water sources.
Value(s) applied	See analysis in the spreadsheets and summary provided in tables 10 and 13 under section B.4 of the PDD.
Choice of data or Measurement methods and procedures	Multiple options were included in an open-ended questionnaire to assess the water sources within the project boundary in line with the methodology guidelines. Data was then analysed in excel spreadsheet to determine the percentage rates.
Purpose of data	Analysis of the baseline water sources
Additional comment	Applies to all project technology types. This has been undertaken at the start of the crediting period and shall be updated in case of a project design change approval where the project boundary is being revised/expanded to new areas. The PP conducted baseline surveys for both households and institutions to determine the baseline water sources. The survey findings are presented in tables 10 and 13 under section B.4 of the PDD.

b. Related to emission reductions.

Data/parameter	SDWS 6: SDG 13 Stove technologies used in the project boundary
Unit	NA
Description	The proportion of different stove types used in premises in the geographical area of the project.
Source of data	The methodology provides that any of the following sources shall be used: - Baseline survey, - Credible published literature for project region, - Studies by academia, NGOs or multilateral institutions, - Official government publications or statistics Source applied must not be more than 3 years old.
Value(s) applied	The PP conducted baseline surveys to collect the data for the parameter and this is provided under the households' baseline surveys and Institutions' baseline surveys spreadsheets tab Baseline fuel consumption. The PP conducted baseline surveys to assess the different stove technologies used for cooking and boiling water by households and institutions within the geographical area of the project. The classification considered the following categories of stove types: - Three-stone fire or a conventional system for woody biomass lacking improved combustion air supply mechanism and flue gas ventilation system. - other conventional systems using woody biomass. - improved cookstoves ($\geq 20\%$ thermal efficiency); and - fossil fuel combusting systems
Choice of data or Measurement methods and procedures	The detailed analysis of the stove technologies and the thermal efficiencies applied is provided in table 16 and 17 under section B.4 of the PDD. Multiple options were included in an open-ended questionnaire to assess the stove technologies used within the project boundary in line with the methodology guidelines. Since the project covers different types of end-users' premises (e.g. households, institutions), the stoves technologies were determined for each premises type.

	Data was then analysed in excel spreadsheet to determine the percentage rates.
Purpose of data	Emission reduction calculations
Additional comment	Applies to all project technology types. This has been undertaken at the start of the crediting period. The PP conducted baseline surveys for both households and institutions to determine the baseline water sources. The survey findings are presented in tables 16 and 17 under section B.4 of the PDD. The sampling criteria outlined in section 4.2.3 of the methodology regarding conducting baseline survey was followed where 198 households and 57 institutions were surveyed.

Data/parameter	SDWS 7: SDG 13
Unit	Expected technical life of project technology
Description	Treatment volume or operational hours or time The expected technical life of an individual project technology shall be defined in the PDD. The details include both technology/device life and filter life if a filter is used and it is replaceable.
Source of data	The methodology provides that for HWT/IWT, any one of the following sources shall be used: - Manufacturer specifications - Third-party certification by a qualified entity, for example recognised certification agency by National/ International Standard body - Commercial guarantee For CWT, any one of the following sources shall be used: - Manufacturer specifications - Guarantee from the installer - Third-party certification by a qualified entity, for example recognised certification agency by National/ International Standard body The PP has adopted use of manufacturer specifications as the source of data for all the filter technologies.
Value(s) applied	The expected technical life of the hollow fibre filters is 5 years and 10 for -butBSF but can be more based on use and maintenance. The filters are hence expected to remain operational all through the first 5-year crediting period. The PD will put in place a maintenance program to ensure any worn out filter membranes (for hollow fibre technology) are replaced so that households can benefit from safe water supply. The maintenance records will be submitted to the GS as supporting evidence.
Choice of data or Measurement methods and procedures	The data for this parameter is sourced from the project fact sheets. The BioSand filters have a lifespan of 10+ years, while the hollow fibre filters have a lifespan of 5+ years with proper use and maintenance. After the 5 years, the PD shall make plans to replace the membranes of the filters that will have worn out so that the households and institutions can continue using the filters.

	The PD will ensure a robust filter monitoring and maintenance plan is in place.
Purpose of data	To estimate the expected technical life of the project technologies.
Additional comment	Applies for all project technology types and has been undertaken at the start of the crediting period. Although the lifespan of the hollow fiber membrane filters is given as 5 years. The lifespan of the filters can go beyond 5 years based on the use and maintenance by end-users. Based on this and given the first crediting period of the project activity is 5 years, there will be no replacement of the filters. However, the PD has a robust technical support team to support the maintenance of the filters on a need basis. In a case where replacement of the membrane is needed, this will be done by the PD at a subsidized cost. <u>as discussed below:</u> i) <u>The project implementer will conduct replacements for the hollow fiber filters which only require replacement of the membrane as the other parts remain intact. The hollow fibre membrane after replacement can serve for another 5 years. The project implementer shall keep updated documentation through the customer complaints form for all replacements indicating the data of the replacement and nature of the replacement. The end-users will sign the document as proof of maintenance, and this will be updated on the sales database. For the BSF filters, continuous maintenance will be done and after the 10-year period, the filters will be reinstalled to enhance their efficiency. The reinstallation basically entails replacement of the sand which is the main filtering medium. This will allow end-users to continue using the filter for a longer period. However, due to geographical spread of the distributed filters and lack of timely communication from the end-users on the operation status of the filters, it will not be possible to conduct all the membrane replacements and hence option ii will be applied in some cases.</u> ii) <u>Upon lapse of the technical lifespan of the filters, all the filters that have reached their useful lifespan from the database, and the membranes will not have been replaced will be removed from the database and no emissions reduction will be claimed from them.</u>

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The lifespan of the membrane filter is dependent on the turbidity of the water and the efficiency of the backwashing process. Turbid water sources tend to reduce the lifespan due to membrane fouling. End-users are advised to report to the project developer on problems such as reduced flow rate and total blockage of the filters to enable timely maintenance of the filters. ~~The PD shall also undertake water quality testing for sampled filters used for more than 5 years and shall advise the end-users on the need to replace their membranes where water quality is found to be wanting.~~

Further, all the filters have a one-year warranty period and the warranty card is signed during the sale of the filter. All repairs within the warranty period are done by the PD for free as per the terms and conditions included in the warranty card which has been provided as supporting evidence.

Data/parameter	SDWS 8: SDG 13
Unit	X _f
Description	Percentage of fuel f use in target population The proportion of each different cooking fuel f used in the project boundary by end-users: - % among the target population if single fuel is used for water boiling. - Weighted average on energy basis, if multifuel situation exists within premise.
Source of data	The methodology provides that any of the following sources shall be used: - baseline survey, - credible published literature for project region, - studies by academia, NGOs or multilateral institutions, - Official government publications or statistics Source applied shall not be more than 3 years old. The PP obtained data for this parameter by conducting baseline surveys for household and institutions end-users. The summary is provided under the Households baseline surveys and Institutions' baseline surveys spreadsheets tab Baseline fuel consumption.
Value(s) applied	The PP conducted baseline surveys to assess the different fuels used by households and institutions in the baseline. The main baseline fuels used included firewood and charcoal as detailed in tables 18 and 22. From the surveys 90.9% of the households were using firewood while 9.1% were using charcoal. These findings are in line with the most recent research on Kenya Household Sector Study conducted by the Ministry of Energy in 2019²⁵, which reports that 93.2% of rural households use wood fuel (fuelwood or charcoal) as their primary fuel (see page 2 of the report). For institutions, 82.5% were using firewood with the rest 17.5% using charcoal. The survey findings also complement the results from the study conducted on use

²⁵ [MoE-2019-Kenya-Cooking-Sector-Study-compressed.pdf \(eedadvisory.com\)](#)

	of biomass cookstoves and fuels in institutions in Kenya by the Clean Cooking Alliance of Kenya (CCAK) in 2018 ²⁶ . The study found that at least 96.6% of the institutions were consuming firewood while 3.4% were using charcoal (See page 6, last paragraph).
Choice of data or Measurement methods and procedures	Multiple options were included in an open-ended questionnaire to assess the fuel used within the project boundary in line with the methodology guidelines. Since the project covers different types of end-users' premises (e.g. households, institutions), the proportion of the fuel used was determined for each premises type.
Purpose of data	Emission reduction calculations
Additional comment	Applies to all project technology types and has been undertaken at the start of the crediting period. The sampling criteria outlined in section 4.2.3 of the methodology regarding conducting baseline survey was followed where 198 households and 57 institutions were surveyed.

²⁶ [2018 10 17 Final Report FA \(ccak.or.ke\)](#)

Data/parameter	SDWS 9: SDG 13 EF _{b,f,CO2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor from use of fuels
Source of data	IPCC defaults
Value(s) applied	For wood and charcoal, the following defaults derived from the IPCC shall be applied: Wood: 112 tCO ₂ /TJ Charcoal: 165.22 tCO ₂ /TJ (includes charcoal production emissions)
Choice of data or Measurement methods and procedures	The current IPCC defaults have been applied.
Purpose of data	Emission reduction calculations
Additional comment	N/A

Data/parameter	SDWS 10: SDG 13 EF _{b,f,nonCO2}
Unit	tCO ₂ e/TJ
Description	Non-CO ₂ emission factor from use of fuels, in case the baseline fuel is biomass or charcoal.
Source of data	IPCC defaults
Value(s) applied	For wood and charcoal, the following defaults derived from the IPCC shall be applied: AR5 GWP - Wood: 9.46 tCO ₂ e/TJ - Charcoal: 44.83 tCO ₂ e/TJ (includes production emissions of CH ₄ and N ₂ O)
Choice of data or Measurement methods and procedures	The current IPCC defaults have been applied.
Purpose of data	Emission reduction calculations
Additional comment	N/A

Data/parameter	SDWS 11: SDG 13 nw _b
Unit	Percentage
Description	Weighted average efficiency of the baseline water boiling devices. Calculate the weighted average of the water boiling efficiency in the project boundary using the proportion of different stove types used and the stove efficiencies.
Source of data	The methodology provides that the following default values shall be applied to calculate the weighted average of the water boiling efficiency in the project boundary: - Three-stone fire or a conventional system for woody biomass lacking improved combustion air supply mechanism and flue gas ventilation system, that is without either a grate or a chimney: default efficiency 10%. - Other conventional systems using woody biomass: default efficiency 20%. - Improved cookstoves: manufacturer specification, or if not available, default efficiency 30%. - Fossil fuel combusting system: manufacturer specification, or if not available, following the testing procedure described in the methodology.
Value(s) applied	The weighted efficiency for households is 11.2 <u>12.4</u> % The weighted efficiency for institutions is 12.3 <u>16.8</u> % Refer to the Households and Institutions baseline surveys spreadsheets tab baseline fuel consumption.
Choice of data or Measurement methods and procedures	Baseline surveys were conducted to identify the different stove technologies used in the project boundary by households and institutions and determine their efficiencies. The detailed analysis is given on tables 16 and 17 under section B.4 of the PDD.
Purpose of data	Emission reduction calculations
Additional comment	The sampling criteria outlined in section 4.2.3 of the methodology regarding conducting baseline survey was followed where 198 households and 57 institutions were surveyed.

Data/parameter	SDWS 12: SDG 13 C _b
Unit	Percentage
Description	Proportion of project end-users who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling.
Source of data	The methodology provides that any of the following sources shall be used: - Baseline survey - Credible published literature for project region - Studies by academia, NGOs or multilateral institutions - Official government publications or statistics Source applied must not be more than 3 years old. The PP conducted baseline surveys to collect data for this parameter for households and institutions end-users.
Value(s) applied	Households: 2.02% Institutions: 3.5% Refer to the Households and Institutions baseline surveys spreadsheets tab baseline water treatment.
Choice of data or Measurement methods and procedures	Baseline surveys were conducted to determine the proportion of households and institutions who in the baseline were already using safe water, either from an improved water source, or from a water treatment method other than boiling. The safe water sources and percentages have been found to be consistent with the information reported for parameter Water sources in the project boundary (SWDS 5).
Purpose of data	Emission reduction calculations
Additional comment	The sampling criteria outlined in section 4.2.3 of the methodology regarding conducting baseline survey was followed where 198 households and 57 institutions were surveyed.

Data/parameter	SDWS 13: SDG 13 q_i
Unit	Litres per hour
Description	Capacity of the household or institutional water treatment technology.
Source of data	The methodology provides that any of the following sources shall be used: - Manufacturer specifications - Design specifications - Third-party certification by a qualified entity, for example recognised certification/ testing agency by National/ International Standard body - Commercial guarantee by the seller An alternative approach can be proposed when information is not available as per the requirements. Professional opinion or expert opinion is not accepted as a source for this parameter. The PP has sourced the data for the parameter from manufacturer specifications.
Value(s) applied	BSF: 60 HHFF: 60 IHFF: 60
Choice of data or Measurement methods and procedures	The manufacturer specifications detail the capacity of the household or institutional water treatment technology. The information is detailed in the filter fact sheets for provided as support document.
Purpose of data	Emission reduction calculations
Additional comment	This applies for household and institutional filters. Where the capacity is expressed in other terms, e.g. litres per day or per minute in the fact sheets, the PP converted this to litres per hour to account for the parameter t_y .

Data/parameter	SDWS 21: SDG 13 $f_{NRB,f,y}$
Unit	Percentage
Description	Fractional non-renewability status of woody biomass fuel during year y , in case the baseline fuel is biomass or charcoal
Source of data	Determined by: - CDM TOOL30, Calculation of the fraction of non-renewable biomass, version 3
Value(s) applied	91.14
Choice of data or Measurement methods and procedures	The PP calculated the $f_{NRB,f,y}$ using the CDMTOOL 30 as indicated in the provided NRB calculation spreadsheet.
Purpose of data	Emission reduction calculation
Additional comment	To quantify the parameter f_{NRB} the PP has selected to use a conservative approach by calculating the value, where a value of 91.14% has been applied instead of using CDM Default Values. The CDM NRB default value of 92%²⁷ was accepted by the CDM EB on 19 September 2012 and was applied as the default value till 18 September 2017 when it expired. The parameter has been fixed ex-ante for the crediting period.

²⁷ [CDM: Fraction of non-renewable biomass \(fNRB\) \(unfccc.int\)](http://unfccc.int)

B.6.3 Ex ante estimation of SDG Impact

SDG 3

Reduced Incidence of the water borne diseases (Skin rash, diarrhoea, foot sore and parasites) and Indoor Air Pollution related diseases/infections (itchy eyes, eye problems and incidence of coughing)

It is assumed that once the households start using the filtered water, there will be a significant reduction in the number of illnesses related to consumption of contaminated water and those associated with indoor air pollution. The project estimates 100% reduction in these illnesses, but this will be monitored during the monitoring surveys.

SDG 6

Water hygiene education campaigns

The project implementor will carry out annual water hygiene education campaigns and the evidence for this shall be provided as part of the monitoring documents. The project proponent shall report the activities conducted each year in the annual monitoring report. Any major changes in the health status of the water users because of contaminated water (e.g., an outbreak of water related disease) shall be reported together with the measured put in place to address it through the hygiene campaign. The impacts of the hygiene campaign shall be assessed using the WHO/UNICEF Joint Monitoring Programme Core questions for drinking water and hygiene to determine the fraction of the households and institutions where Safe water and Hygiene practices are found to fulfil "safely managed" or "basic" requirements.

The WHO/UNICEF Joint Monitoring Programme Core questions for drinking water and hygiene shall be applied as summarised below:

A) Core questions on water and hygiene for household surveys

Service type	Core questions	New JMP service ladders	
Drinking water			
1. Safely managed services	W3	SAFELY MANAGED	Drinking water from an improved water source that is located on premises, available when needed and free from faecal and priority chemical contamination
1a – accessibility	W5		
1b – availability	W6		
1c – quality			

2. Basic and limited services	W2, W4	BASIC	Drinking water from an improved source, provided collection time is not more than 30 minutes for a round trip, including queuing
		LIMITED	Drinking water from an improved source for which collection time exceeds 30 minutes for a round trip, including queuing
1. Improved or unimproved; surface water	W1	UNIMPROVED	Drinking water from an unprotected dug well or unprotected spring
		SURFACE WATER	Drinking water directly from a river, dam, lake, pond, stream, canal, or irrigation canal
Hygiene			
1. Facility or no facility	H1	BASIC	Availability of a handwashing facility on premises with soap and water.
2. Basic and limited handwashing facility	H2, H3		
		LIMITED	Availability of handwashing facility on premises without soap and water.

		NO FACILITY	No handwashing facility on premises.
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B) Core questions on water and hygiene for Schools surveys

Service type	Core questions	New JMP service ladders	
Drinking water			
1. Advanced service	W1 W2	Advanced	To be defined at national level
2. Basic services		Basic	Drinking water from an improved source and water is available at the school at the time of the survey
3. Limited service		Limited	Drinking water from an improved source but water is unavailable at the school at the time of the survey
4. No service		No service	Drinking water from an unimproved source or no water source at the school
Hygiene			
1. Advanced service	H1	Advanced	Additional criteria may include hygiene education, group handwashing, menstrual hygiene materials, and

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	H2		accessibility to all users.
2. Basic service		Basic	Handwashing facilities with water and soap available at the school at the time of the survey.
2. Limited service		Limited	Handwashing facilities with water but no soap available at the school at the time of the survey.
3. No service		No service	No handwashing facilities available or no water available at the school.

C) Core questions on water and hygiene in health care facilities

Service type	Core questions	JMP service ladders	
Drinking water			
1. Advanced service	G-W1 G-W2 G-W3	Advanced	To be defined at national level
2. Basic services		Basic	Water is available from an improved source located on premises.
3. Limited service		Limited	An improved water source is within 500 meters of the facility, but not all requirements for basic service are met
4. No service		No service	Water is taken from unprotected dug wells or springs, or surface water sources; or an improved source that is more than 500 m from the facility; or the facility has no water source.
Hygiene			
1. Advanced service	G-H1 G-H2	Advanced	To be defined at national level
2. Basic service		Basic	Functional hand hygiene facilities (with water and soap and/or alcohol based hand rub) are available at points of care, and

			within 5 meters of toilets.
3. Limited service		Limited	Functional hand hygiene facilities are available at either points of care or toilets, but not both.
4. No service		No service	No functional hand hygiene facilities are available at either points of care or toilets

SDG 6

Number of people attending trainings, workshops and/or seminars organised on filter installation and maintenance.

The project will organise relevant trainings and workshops on filter assembly, installation, use and maintenance for the filter installers to equip them with the required knowledge and skills. A value of 30 people has been estimated for the training opportunities.

SDG 8

Number of persons employed by the project.

It is anticipated that the project will create both permanent and temporary job opportunities to the local people in the respective project areas. These opportunities are expected to arise during the installation, distribution, and maintenance of the filters, during the annual monitoring surveys, and overall management of the program. This parameter will be determined by reviewing the employment records of the project to determine number of people employed by gender and age, and their levels of income. A value of 50 people has been estimated for the formal employment opportunities.

SDG 13

Emission reductions

The detailed calculation method is shown in section B.6.1 of this PDD. The transparent ex-ante calculations of the outcome for SDG 13 (GHG emission reductions) are provided in a provided Excel Spreadsheet uploaded to GS Registry. For data/parameters available at the time of design certification, values contained in section B.6.2 of this PDD and for data/parameters only available after monitoring the estimates contained in section B.7.1 of this PDD have been used. The outcomes for SDG 13 for each technology are shown in the following tables.

HHHF	
5 year emission reduction estimations	
Crediting period	Emission reductions
2022	<u>9317.51</u> 1813.3
2023	<u>26414.63</u> 3489.9
2024	<u>60272.17</u> 6416.4
2025	<u>161822.32</u> 05167.6
2026	<u>223185.82</u> 82967.7
2027	<u>24749.83</u> 1379.2
Total	<u>505,762.00</u> 641,234.00
Average for 5 years	<u>101,152.00</u> 128,246.00

BSF	
5 year emission reduction estimations	
Crediting period	Emission reductions
2022	64.782.1
2023	268.8340.8
2024	532.0674.5
2025	950.91205.6
2026	1399.71774.6
2027	159.7202.4
Total	3,375.00 4,279.00
Average for 5 years	675.00 855.00

IHFF	
5 year emission reduction estimations	
Crediting period	Emission reductions
2022	4739.46008.9
2023	11871.015050.8
2024	23182.429391.9
2025	50550.564090.8
2026	77863.798720.0
2027	8661.310981.3
Total	176,868.00 224,243.00
Average for 5 years	35,373.00 44,848.00

CWT	
5 year emission reduction estimations	
Crediting period	Emission reductions
2022	4819.26582.3
2023	13620.618603.8
2024	24896.634005.1
2025	40475.155283.1
2026	43629.559591.5

2027	<u>4834.36602.9</u>
Total	<u>132,275.00</u> 180,668.00
Average for 5 years	<u>26,455.00</u> 36,133.00

Combined Emission Reductions for the 5 years for all technologies

Year	BE _y (tCO ₂ e)	PE _y (tCO ₂ e)	LE _y (tCO ₂ e)	Emission Reductions (ER _y , tCO ₂ e)
2022	<u>18,940.90</u> <u>24,486.62</u>	0	0	<u>18,941</u> <u>24,487</u>
2023	<u>52,175.00</u> <u>67,485.22</u>	0	0	<u>52,175</u> <u>67,485</u>
2024	<u>108883.11</u> <u>40488</u> <u>-0</u>	0	0	<u>108,883</u> <u>140,488</u>
2025	<u>253798.93</u> <u>25747</u> <u>-1</u>	0	0	<u>253,799</u> <u>325,747</u>
2026	<u>346078.74</u> <u>43053</u> <u>-8</u>	0	0	<u>346,079</u> <u>443,054</u>
2027	<u>38405.14</u> <u>9165.9</u>	0	0	<u>38,405</u> <u>49,166</u>
Totals	<u>818,281.00</u> 1,050,426.00	0	0	<u>818,281</u> 1,050,426
Annual average	<u>163,656.00</u> 210,085.00	0	0	<u>163,656</u> 210,085

B.6.4 Summary of ex ante estimates of each SDG Impact

The detailed calculation method is shown in section B.6.1 of this PDD. The transparent ex-ante calculations of the outcome for SDG 13 (GHG emission reductions) are provided in a separate Excel Spreadsheet uploaded to GS Registry. The summary below is the combined ER estimates for the project technologies included in the project namely BSF, HHHF, IHFF and CF.

SDG 13

Year	Baseline estimate	Project estimate	Net benefit
09/02/2022	<u>18,940.90</u> 24,486.62	0	<u>18,940.90</u> 24,486.62
2023	<u>52,175.00</u> 67,485.22	0	<u>52,175.00</u> 67,485.22
2024	<u>108883.11</u> 40488.0	0	<u>108883.11</u> 40488.0
2025	<u>253798.93</u> 25747.1	0	<u>253798.93</u> 25747.1
2026	<u>346078.74</u> 43053.8	0	<u>346078.74</u> 43053.8
08/02/2027	<u>38405.14</u> 9165.9	0	<u>38405.14</u> 9165.9
	<u>818,281.00</u> 1,050,426.00	0	<u>818,281.00</u> 1,050,426.00
Total			
Total number of crediting years	5		5
Annual average over the crediting period	<u>163,656.00</u> 210,085.00		<u>163,656.00</u> 210,085.00

Formatted Table

SDG 8 (number of people)		
Year	Number of people employed by the project	Percentage of employees with salaries paid at par with the average wage of Kenya
<u>09/02/2022</u>	25 people (15 males, 10 females)	100%
2023	25 people (15 males, 10 females)	100%
2024	30 people (15 males, 15 females)	100%
2025	30 people (15 males, 15 females)	100%
2026	50 people (25 males, 25 females)	100%
<u>08/02/2027</u>	50 people (25 males, 25 females)	100%

Total	50 people (25 males, 25 females)	100%
Average	50 people (25 males, 25 females)	100%

SDG 6 (number of people)	
Year	Number of people attending water hygiene education campaigns.
<u>09/02/2022</u>	100
2023	100
2024	100
2025	100
2026	100
<u>08/02/2027</u>	50
Total	550
Average	110

SDG 6 (number of people)	
Year	Number of people attending training/ workshops on filter installation and maintenance
<u>09/02/2022</u>	30
2023	30
2024	30
2025	30
2026	30
<u>08/02/2027</u>	10
Total	160
Average	32

SDG 3 (percentage)	
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Year	Reduction in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases
09/02/2022	100%
2023	100%
2024	100%
2025	100%
2026	100%
08/02/2027	100%
Total	100%
Average	100%

SDG 3 (percentage)	
Year	Improvement in Indoor Air Quality
09/02/2022	100%
2023	100%
2024	100%
2025	100%
2026	100%
08/02/2027	100%
Total	100%
Average	100%

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored.

a. Parameters related to water quality.

Data / Parameter	SDWS18: SDG 6
Unit	Mq,y
Description	Fraction
	Ongoing water quality indicated as the fraction of the samples that pass microbial quality standard

	requirements specified in relevant microbial quality standard for drinking water of the host country. In case a national standard is not available, the water quality shall comply with WHO Guideline values for verification of microbial quality i.e., all water directly intended for drinking must not have detectable <i>E. coli</i> in any 100 ml sample i.e., less than 1 Colony Forming Unit (CFU) of <i>E. coli</i> /100 ml.
Source of data	Water Quality Testing. Testing of water all appliances or at a representative sample of end-users at the following point where the water reaches the end-user premises: HWT and IWT: testing of the water that exits the treatment technology. CWT: where water is supplied directly to the premises of the household or institution (e.g. piped): testing of the water entering the household (e.g. coming out of the tap). CWT: where water is retrieved from the CWT or CWS location: testing of the water in the transport containers when it reaches the end-user premises (e.g. household, institution). The water quality test applies the bacterial quality standard
Value(s) applied	95%
Measurement methods and procedures	The project shall make use of the 3M™ Petrifilm™ <i>E. coli</i>/Coliform Count Plates²⁸. The specific description of the kits are as follows: - These plates provide both <i>E. coli</i> and total Coliform Count information with confirmed results in just 24-48 hours. By eliminating subsequent confirmation steps required with most traditional reference methods, the 3M Petrifilm <i>E. coli</i>/Coliform Count Plates can help increase productivity and reduce overall lab costs. - Plates are sample-ready, eliminating the time-consuming, cumbersome step of preparing media/agar dishes.

²⁸ [3M™ Petrifilm™ E. coli/Coliform Count Plates | 3M United States](#)

	- Can be used with the 3M™ Petrifilm™ Plate Reader for consistent, fast, automated reading and recording of results. - This plate has been awarded Official Methods of Analysis OMA #991.14.for Food OMA # 998.08 for Poultry, Meat and Sea Food. The 3M Petrifilm Plates have demonstrated consistent performance that has been cited globally with more than 200 certificates, recognitions, validations, and peer-review publications. For more than 30 years, food safety professionals around the world have put their trust in 3M Petrifilm Plates. That's because 3M Petrifilm Plates have streamlined, standardized, and simplified the process of microbial quantitative indicator testing, improving productivity and helping ensure the highest levels of product quality.
Monitoring frequency	Annual and the first round of testing shall be conducted at least after six months from the start date.
QA/QC procedures	Field testing kits based on Colony Forming Unit method or Most Probable Number method shall be used. To use the field testing kits the project shall meet the following requirements: - Testing kits must be approved by national agency or meet standards set by relevant international organisation e.g. US-EPA, and - Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring, whereby local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits.
Purpose of data	Emission reduction calculations
Additional comment	If the proportion of samples not meeting Safe Drinking Water Quality Standards exceeds a threshold, no emission reductions can be claimed for the corresponding monitoring period. Thresholds: - Project or VPA year 1: 20% - Project or VPA year 2: 15% - Project or VPA year 3 or above: 10%. The sampling results shall satisfy at minimum the 90/10 rule, i.e., the endpoints of the 90% confidence interval lie within +/- 10% of the estimated proportion in

relative units. A minimum sample of size 30 shall be applied.
Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 20: SDG 6 Water hygiene education campaigns
Unit	NA
Description	Hygiene campaigns carried out among project safe water end-users.
Source of data	Report of annual hygiene campaigns results
Value(s) applied	90%
Measurement methods and procedures	The impacts of the hygiene campaign shall be assessed using the WHO/UNICEF Joint Monitoring Programme Core questions for drinking water and hygiene to determine the fraction of the households and institutions where Safe water and Hygiene practices are found to fulfil "safely managed" or "basic" requirements. In-person or telephone or by messaging (e.g. text, app) based survey shall be conducted covering all the JMP core questions for drinking water and core questions for hygiene. The fraction of the households where Safe water and Hygiene practices are found to fulfil "safely managed" or "basic" requirements is expected to increase over time as a result of the hygiene campaigns.
Monitoring frequency	Annual
QA/QC procedures	The detailed method that shall be used to assess hygienic handling of clean water for households, schools and health centres is provided in section B.6.1 of the PDD and shall be verified by the VVB.
Purpose of data	Hygiene survey assessment
Additional comment	The project developer shall report the activities conducted each year in a detailed "Report of annual hygiene campaigns results" and summarize the results in the project monitoring reports. Any major changes in the health status of the water users as a result of contaminated water (e.g. an outbreak of water related disease) must be reported and, if relevant, a strategy put in place to address it through the subsequent hygiene campaign. Applies to all project technology types. Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters. In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied.

	Value applied is an estimate for purposes of ex-ante estimation, actual value to be determined during monitoring.
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b. Parameters related to SDG outcomes

Data / Parameter	SDWS 19: SDG 3
	Reduction in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases.
Unit	Percentage
Description	Percentage of the population that experienced a decrease in incidences of waterborne diseases such as skin rashes, diarrhoea, foot sores, parasites, eye problems and other waterborne diseases because of using the water filters.
Source of data	Annual sustainable development surveys
Value(s) applied	100%
Measurement methods and procedures	Multiple options in an open-ended questionnaire to eliminate any possibility of bias. Data will then be analysed in excel spreadsheet to determine the percentage rates.
Monitoring frequency	Annual
QA/QC procedures	Trained data enumerators will be charged with the responsibility of collecting data.
Purpose of data	Sustainable development outcomes assessment
Additional comment	Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters. In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied. Value applied is an estimate for purposes of ex-ante estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 19: SDG 3
Unit	Improvement in Indoor Air Quality
Description	Percentage
Source of data	Percentage of population that have experienced improvement smoke levels, incidence of coughing, respiratory illness, and itchy eyes.
Value(s) applied	Annual sustainable development surveys
Measurement methods and procedures	100%
Monitoring frequency	Multiple options in an open-ended questionnaire to eliminate any possibility of bias. Data will then be analysed in excel spreadsheet to determine the percentage rates.
QA/QC procedures	Annual
Purpose of data	Trained data enumerators will be charged with the responsibility of collecting data.
Additional comment	Sustainable development outcomes assessment
	Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters. In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied.
	Value applied is an estimate for purposes of ex-ante estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 19: SDG 8
Unit	Number of people employed by the project
Description	Number of people This parameter will be collected to indicate the project provides decent work for all men and women.
Source of data	Employment records
Value(s) applied	50
Measurement methods and procedures	The number will be derived from the employment records which will be kept and updated by the project developer.
Monitoring frequency	Annual
QA/QC procedures	PD shall submit employment records as a support document.
Purpose of data	Sustainable development outcomes assessment
Additional comment	The project shall employ both men and women to ensure equality in opportunities that arise. Value applied is an estimate for purposes of ex-ante estimation, actual value to be determined during monitoring.

c. Related to emission reductions

Data / Parameter	SDWS22: SDG 6 X_{cleanboil,y}
Unit	Percentage
Description	Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year y.
Source of data	Project survey This survey may be performed in person, by telephone, by messaging (e.g., text, app), appropriate to the context.
Value(s) applied	0
Measurement methods and procedures	A question will be included in the survey questionnaire to assess the proportion of end-users boiling safe water. Data will then be analysed in excel spreadsheet to determine the percentage.
Monitoring frequency	Annually
QA/QC procedures	The project developer shall ensure accurate and transparent record keeping, monitoring and evaluation of installed units and safe water purchases. Electronic data records shall be used to store data from the Field Tests and surveys. All electronic records shall be stored up to 2 years after the project activity ends and shall be made accessible for spot checking and cross referencing by a third party.
Purpose of data	Emission reduction calculations
Additional comment	The minimum sample size for shall be determined considering the project technology type and in line with the sampling approach applied. In all cases the recommended minimum sample of 30 shall be maintained. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS23: SDG 6 $Q_{m,y}$
Unit	Liters/year
Description	Monitored quantity of safe water provided by the CWT project in year y
Source of data	At the central location of the CWT: Option 1: Flow meter measures water volume directly
Value(s) applied	10,000
Measurement methods and procedures	The water volume will be derived from the flow meter readings.
Monitoring frequency	Continuous
QA/QC procedures	Follow manufacturer, sector, national or international standards or guidelines for calibration and maintenance of the measurement device
Purpose of data	Emission reduction calculations.
Additional comment	Applies to Community water filters. Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved where the parameter is sampled. A minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS24: SDG 6 QPW_p
Unit	Liters/person/day
Description	Volume of drinking water per person per day for premises type
Source of data	Option 2: Water Consumption Field Tests. - In all cases, the value is capped at 5.5 L/person/day.
Value(s) applied	5.5
Measurement methods and procedures	The water consumption field test (WCFT) will measure project-supplied clean water consumption volumes. The WCFT will be conducted with end-user's representative of the project scenario target population and currently using the project technology. The WCFT shall be conducted over 3 days, not including weekends, and averaged value (l/person/day) value should be determined after excluding outliers.
Monitoring frequency	Biennial
QA/QC procedures	The WCFT shall be designed to ensure that monitoring is representative of typical technology use practices and that: - it is transparent and can easily be replicated, - it is evidently conservative, - the sample is randomly selected so as to not introduce a material bias, and - the impact of daily and seasonal variations on the expected average water consumption is accounted for. Follow manufacturer, sector, national or international standards or guidelines for calibration and maintenance of the measurement device.
Purpose of data	Emission reduction calculations
Additional comment	An appropriate sampling method shall be used and shall ensure the sample is selected randomly. The minimum sample size for shall be determined considering the project technology type and in line with the sampling approach applied. In all cases the recommended minimum sample of 30 shall be maintained. Applies for all filter technologies. Values applied are estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 25: SDG 6 HN_{p,y}
Unit	Number
Description	Number of individuals per premises type p in the project boundary in year y
Source of data	Project survey or (Official government publications or statistics; Credible published literature for project region, or Studies by academia, NGOs or multilateral institutions)
Value(s) applied	BSF&HHFF: 5.15 IHFF: 450
Measurement methods and procedures	A question will be included in the survey questionnaire to assess the number of people per premises. Data will then be analysed in excel spreadsheet to determine the average.
Monitoring frequency	Annual
QA/QC procedures	The value applied shall be cross-checked against at least one other source on the list. For cross-check purposes, sources applied may be up to 5 years old.
Purpose of data	Emission reduction calculations
Additional comment	For project surveys, the minimum 90% confidence interval and a 10% margin of error requirement shall be achieved. A minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied. Value applied is derived from the baseline surveys and are applied for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 26: SDG 6 HH _{p,y}
Unit	Number
Description	Number of premises type p served by the project in year y
Source of data	Survey of the premises (e.g., households, schools) within 1 km distance of project water source to check how often the premises used the project water source during the year. This survey may be part of the project survey and may be performed in person, by telephone, by messaging (e.g. text, app), appropriate to the context. Premises that report at least every-two-days use may be counted.
Value(s) applied	To be determined
Measurement methods and procedures	A question will be included in the survey questionnaire to assess the number of premises served or the number of people served in case of CWT installed in institutions. Data will then be analysed in excel spreadsheet to determine the total number.
Monitoring frequency	Annually
QA/QC procedures	The population of each institution where Community filters are installed will be recorded during installation of the filters and will be cross checked during monitoring.
Purpose of data	Emission reduction calculations
Additional comment	The minimum 90% confidence interval and a 10% margin of error requirement shall be achieved. A minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied. Applies to community water filters. Where the community filters are installed in institutions, the description of this parameter will be assumed to be the number of people served by the community filter.

Data / Parameter	SDWS 27: SDG 6 $DO_{p,y}$
Unit	Days
Description	Days the project technology is operational for end-users in premises p in year y.
Source of data	In order of preference: 1. Measure directly using operation sensor, or 2. Demonstrate from log of operation and maintenance system.
Value(s) applied	347
Measurement methods and procedures	The number of days for schools and other institutions, served by institutional and community filters shall be limited by the number of school days in the period, taking into account weekends and holidays as applicable.
Monitoring frequency	Annually
QA/QC procedures	Values higher than 347 days may only be applied when option 1 is used. 347 days is 95% of days, in line with pump-maintenance in the literature. For schools and other institutions, as applicable, the days shall also be limited by the number of school days in the period, taking into account weekends and holidays.
Purpose of data	Emission reduction calculations
Additional comment	Applies to institutional and community water filters. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 28: SDG 13 $N_{p,y}$
Unit	Number
Description	Accumulated number of premises type p with at least one individual project technology in year y
Source of data	Sales records
Value(s) applied	BSF: 500 HHFF: 77,500 IHFF: 1,300 CF: 120
Measurement methods and procedures	The number of premises and technologies supplied will be derived from the sales database.
Monitoring frequency	Annually
QA/QC procedures	The project developer shall maintain updated sales records which shall include: • Date of sale/distribution • Geographic area of sale • Model/type of project technology sold. • Quantity of project technologies sold. Name and telephone number, and address (if available) or other traceable indicator of premises identity and location for all end users.
Purpose of data	Emission reduction calculations
Additional comment	Applies to household and institutional water filters. Units shall not be counted in $N_{p,y}$ after the end of their technical life, unless this is addressed by the measures to manage the cases where the expected technical life of the project technology is shorter than the crediting period, namely replacement or retrofit. Value applied is a projected estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 29: SDG 13 Up,y
Unit	Percentage
Description	Usage rate of the project technology by premises type p during year y
Source of data	Survey the premises with a project technology to determine the usage rate of the project technology during the year. The following usage survey guidelines shall be followed: Option 1: In-person survey of project premises (e.g. households, schools) covering all topics outlined in Annex - 1. Households that show at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 100% to get Up,y. Option 2: Survey performed by telephone or messaging (e.g. text, app), covering topics 1, 2 and 3 of Annex -1. Households that report at least once-in-two-days use may be counted as users. The resulting fraction is multiplied by 75% to get Up,y
Value(s) applied	95
Measurement methods and procedures	The data will be derived from the usage survey which will follow the usage survey criteria set out in the methodology. The PP intends to use the Usage survey questionnaire developed by the GS for conducting the survey.
Monitoring frequency	Annually
QA/QC procedures	Where a WCFT is undertaken to determine QPWp, this may be used to cross check the usage percentage.
Purpose of data	Emission reduction calculation
Additional comment	Applies to all HWT and IWT technologies and projects. Where project technologies of different ages are being credited, the sample shall be representative of the distribution of project technology ages. The minimum sample size for HWT - for individual technology age-group shall be minimum 30 household. The minimum sample size for IWT – for individual technology age group shall be determined considering the project technology type and in line with the sampling approach applied. For HWT, the Usage Surveys Guidelines outlined under annex 1 of the methodology shall be applied. PP shall follow either of the two options given to conduct the

	survey. As required under the guidelines, sufficient precautions shall be taken by enumerators to ensure that there is no bias while conducting the usage surveys. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.
Data / Parameter	SDWS 30: SDG 13
Unit	$t_{p,y}$
Description	Hours per day
Source of data	Usage time of the project technology by premises type p in year y
Value(s) applied	Determined via Project survey
Measurement methods and procedures	Option 1. Observational sample-based survey of project household practices.
Monitoring frequency	Option 2. Interview survey performed by telephone or messaging (e.g. text, app).
QA/QC procedures	Option 3. Default of 5 hours.
Purpose of data	5
Additional comment	A question will be included in the survey questionnaire to assess the number of hours per day the technologies are used in the premises. Data collected will then be analysed in excel spreadsheet to determine the average hours per day.
	Annual
	N/A
	Emission reduction calculations
	Applies to household and institutional filters. The default value of 5 hours is applied for purposes of ex-ante ER estimation, actual value to be determined during monitoring.
	For project surveys, the minimum 90% confidence interval and a 10% margin of error requirement shall be achieved. A minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied.

Data / Parameter	SDWS 31: SDG 13 $DP_{p,y}$
Unit	Days
Description	Average days the project technology is present for end-users in the premises p in year y
Source of data	Sales records
Value(s) applied	Households: 365 Institutions: 347
Measurement methods and procedures	Based on the sales or distribution records of "Date of sale/distribution" and ex-ante parameter "Expected technical life of project technology," how many days of the 365 days of the year it was in the premises and within its technical life shall be determine for each project device. The average for all the project technology by premises type p will then be calculated to obtain this parameter.
Monitoring frequency	Annually
QA/QC procedures	For schools and other institutions, as applicable, the days shall be limited by the number of school days in the period, taking into account weekends and holidays.
Purpose of data	Emission reduction calculations
Additional comment	Applies to household and institutional filters. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 32: SDG 13 DNp,y
Unit	Number
Description	Average number of individual project technologies in each project premises type p in year y.
Source of data	Sales records
Value(s) applied	Households: 1 Institutions: 2
Measurement methods and procedures	Based on the sales or distribution records of "Quantity of project technologies sold" and identifying information of buyer/recipient, the average number of project devices per premises will be calculated. The average number shall then be determined per premises type p. (e.g. households, institutions).
Monitoring frequency	Annually
QA/QC procedures	The project developer shall maintain an updated sales database for all project technology types.
Purpose of data	Emission reduction calculations
Additional comment	Applies to household and institutional filters. An average of 2 filters is applied for institutions since most of them prefer to have more than one filter to meet the water demand for the population. Value applied is an estimate for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

Data / Parameter	SDWS 32: SDG 13 LEp,y
Unit	tCO2e per year
Description	Leakage emissions during year y
Source of data	Sources established by following Leakage emissions section
Value(s) applied	0
Measurement methods and procedures	Monitoring parameters required for calculating leakage emissions shall be included in the monitoring plan in the PDD as required to monitor and quantify the sources of leakage determined by following the Leakage emissions section.
Monitoring frequency	Biennially
QA/QC procedures	Compliance with the general requirements for sampling and general requirements for data and information sources.
Purpose of data	Emission reduction calculation
Additional comment	Applies to household and institutional filters. An average of 2 filters is applied for institutions since most of them prefer to have more than one filter to meet the water demand for the population. The leakage assessment is detailed under section B.7.3 of the PDD. The value applied is from this assessment and has been applied for purposes of ex-ante ER estimation, actual value to be determined during monitoring.

B.7.2 Sampling plan

Sampling Method

The sampling method chosen for the project activity is Stratified Random Sampling. This will be done in accordance with the Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 4, and has been found to be suitable since the project activity shall distribute different filter technologies and has two baseline scenarios i.e households and institutions. The project proponent shall maintain an updated Total sales Record (TSR) for each technology type which will identify the various geographical zones of operational Filter's technologies within the project activity. Random test subjects shall hence be identified from the TSR. Random sampling will be done independently for each technology type which will form the strata. These will include the Household Biosand (BSF), Household Hollow Fibre Filters (HHHF), Institutional Hollow Fibre filters (IHFF) and Community filters (CF) to allow representation of the two baseline scenarios.

Objectives

Objectives of sampling is to accurately quantify the usage rate, water consumption characteristics in the project scenario, sustainable development impacts and dropout rate of the project technologies distributed to households.

Sample Size:

In accordance with general requirements for sampling stipulated in Section 4.2 of the methodology, the PP shall determine sample sizes using the applicable precision level for the different parameters to be monitored by the project.

A statistically valid sample can be used to determine parameter values, as per the relevant requirements for sampling in the "Methodology for Sampling and surveys for CDM project activities and programme of activities." Minimum 90% confidence interval and a 10% margin of error requirement shall be achieved for the sampled parameters.

In any case, for proportion parameter values, a minimum sample size of 30, or the whole group size if this is lower than 30, must always be applied.

When a baseline and project survey is used the following sample size guidelines should be applied, unless otherwise stated for specific parameters:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller.
- Group size 300 to 1000: Minimum sample size 10% of group size
- Group size > 1000 Minimum sample size 100

All end-users included in surveys or sampling related to baseline determination or operation of the project shall be located within the 1 km radius of the CWT location.

The following steps shall be applied in determining the sample size:

1. List all the filters distributed vintage wise and taking into account the different filter technologies and geographical regions (counties) where filters have been distributed.
2. Calculate a cumulative sum of all the filters sold, where the last number in this series shall be the total number of filters sold.
3. Determine a sample size of a minimum 30 households from each vintage year.
4. The total number of filters shall then be divided by the sample size (30). The result shall be used to determine the Sample Interval.
5. At random a number between 1 and the Sample Interval (SI) shall be chosen this shall be held as the Random Start (RS)
6. Sample units (SU) shall be identified as follows:
 - a. $SU1 = RS$
 - b. $SU2 = RS + SI$
 - c. $SU3 = RS + 2SI$
 - d. $SUN = RS + (n-1)SI$
7. These sample units (SI) correspond to an identified household in the cumulative sales in the Sales Record.

B.7.3 Other elements of monitoring plan

1. Monitoring Procedure

A total sales record and a project database shall be maintained continuously and stored on an electronic database by the project developer. This will then be shared by the carbon asset developer for cross checking. A project and usage survey will be conducted annually and as prescribed in the methodology. The monitoring exercise shall be a combined function of the usage survey, project survey and the Water

Consumption Field Test exercises. The monitoring exercise shall be carried out separately for each technology type – namely the BSF and the Hollow Membrane Fibre Units for households, institutions, and community filters.

A. Total Sales Record

The project proponent will maintain an accurate and complete sales record. The record will be backed up electronically by both the project developer and ClimateCare in its capacity as the carbon asset development consultant. The required data fields are:

1. Date of sale/distribution
2. Geographic area of sale
3. Model/type of project technology sold
4. Quantity of project technologies sold
5. Name and telephone number and address (where available)
6. Mode of use: domestic, commercial or other
7. Unique Filter serial number

B. Project Database

The project database will be derived from the total sales record. The differentiation of the project database into sections is based on the results of the applicable monitoring studies for each project scenario, in order that ER calculations can be conducted appropriately section by section. Technologies aged beyond their useful lifetime, as established in the usage survey, shall be removed from the project database, and no longer credited.

2. On-going Monitoring Studies

a) Monitoring Survey – *Completed annually, beginning 1 year after project registration.*

The project participant will conduct annual monitoring surveys which will investigate annual changes in the project scenario.

Parameters for assessment include:

1. Usage rate in project scenario during Year X
2. Number of premises served by the project.
3. Number of individuals per premises.
4. Days the project technology is operational for end-user.
5. Accumulated number of premises.

6. Usage time of the project technology.
7. Average days the project technology is present for end-users.
8. Average days the project technology is present for end-users.
9. Volume of drinking water per person per day.
10. Proportion of project end-users that boil safe (treated, or from safe supply) water after installation of project technology in year.
11. Ongoing water quality.
12. Monitored quantity of safe water provided by the CWT.
13. Water hygiene education campaigns

b) Usage Survey – Completed annually

The usage survey provides a single usage parameter that is weighted based on drop off rates that are representative of the age distribution for project technologies in the total sales record.

The usage parameter in the usage survey shall be based on users who completely abandon use of the Aqua Clara filters and those that do not properly use or maintain the filter to affect the quality of water treated. This survey will be carried out using closed questionnaires and spot checks.

The minimum total sample size will be 100, with at least 30 samples for project technologies of each age being credited.

c) Project Field Test Update – Completed annually

The Project Field Tests update is an extension of the project PFT and provides a water consumption assessment representative of project technologies currently every two years. Hence the PFT update accounts for changes in the project scenario over time as project technologies age and new customers are added, also as new models and designs are introduced.

d) Water Quality Tests– Completed annually

The methodological requirement is that water quality must be tested annually, with the first round of testing within 6 months of the stated project start date. On the water quality testing

1. Laboratories used for water quality testing must be approved by local health authorities and/or have quality accreditation; and

2. The laboratory used must demonstrate that it has an adequate quality management plan in place which addresses both quality assurance and quality control test procedures.

3. Field testing kits also are eligible, e.g., based on Colony Forming Unit method or Most Probable Number method.

To use the field testing kits the project shall meet the following requirements:

a. Testing kits must be approved by national agency or meet standards set by relevant international organisation e.g. US-EPA.

The project intends to make use of the 3M™ Petrifilm™ E. coli/Coliform Count Plates²⁹.

The specific description of the kits are as follows:

- These plates provide both *E. coli* and total Coliform Count information with confirmed results in just 24-48 hours. By eliminating subsequent confirmation steps required with most traditional reference methods, the 3M Petrifilm *E. coli*/Coliform Count Plates can help increase productivity and reduce overall lab costs.
- Plates are sample-ready, eliminating the time-consuming, cumbersome step of preparing media/agar dishes.
- Can be used with the 3M™ Petrifilm™ Plate Reader for consistent, fast, automated reading and recording of results.
- This plate has been awarded Official Methods of Analysis OMA #991.14.for Food OMA # 998.08 for Poultry, Meat and Sea Food

b. Testing kits shall be tested for its accuracy and robustness prior to application for project level monitoring, whereby local or accredited laboratory shall conduct water quality tests using testing kits and a relevant ISO standard or an equivalent standard, in parallel with field testing kits.

The 3M Petrifilm Plates have demonstrated consistent performance that has been cited globally with more than 200 certificates, recognitions, validations, and peer-review publications. For more than 30 years, food safety professionals around the

²⁹ [3M™ Petrifilm™ E. coli/Coliform Count Plates | 3M United States](#)

world have put their trust in 3M Petrifilm Plates. That's because 3M Petrifilm Plates have streamlined, standardized, and simplified the process of microbial quantitative indicator testing, improving productivity and helping ensure the highest levels of product quality.

The PP conducted the required water quality testing for households, institutions and community filters using the 3M™ Petrifilm™ E. coli/Coliform Count Plates. The water quality testing was conducted between 03/05/2021 and 14/07/2021.

The PP collected samples of water from 70 households, 30 institutions and 8 community filters which had been randomly selected as sampling units from the technology in the sampling frame. Each water sample was labelled and stored in cool packs. At a central field laboratory, the samples were tested using the 3M™ Petrifilm™ E. coli/Coliform Count Plates., and results recorded for each individual sample. Field laboratory testing has been selected because it allows for near immediate testing without contamination or any other distortion from the transportation to laboratories.

The sample size required for water quality testing using testing kits was as per the sample size requirement for baseline surveys as given in applicable methodology i.e.

- Population size <300: Minimum sample size 30 or population size, whichever is smaller.
- Population size 300 to 1000: Minimum sample size 10% of group size
- Population size > 1000 Minimum sample size 100

The total sample size for the water quality testing was 108 including 70 households, 30 institutions and 8 community filters.

i) Water quality testing procedure

Water quality testing shall be conducted by use of the 3M™ Petrifilm™ E. coli/Coliform Count Plates.

How the 3M™ Petrifilm™ E. coli/Coliform Count Plates works

1. The 3M™ Petrifilm™ E. coli/Coliform Count Plates provide a cost-effective, convenient, and reliable method for testing equipment, raw materials, food products and manufacturing environmental samples.
2. The 3M Petrifilm E. coli/Coliform Count Plate is a sample-ready-culture-medium system which contains Violet Red Bile (VRB) nutrients, a cold-water-soluble gelling agent, an indicator of glucuronidase activity (BCIG), and a tetrazolium indicator that facilitates colony enumeration in food and beverage samples.
3. These plates provide both E. coli and total Coliform Count information with confirmed results in just 24-48 hours. By eliminating subsequent confirmation steps required with most traditional reference methods, the 3M Petrifilm E. coli/Coliform Count Plates can help increase productivity and reduce overall lab costs.
4. Three Easy Steps Offer Fast, Easy-to-Interpret Results:
 - i. Inoculate - 3M Petrifilm E. coli/Coliform Count Plates are easy to inoculate. Lift the top film and add sample.
 - ii. Incubate - The space saving design minimizes incubator space requirements.
 - iii. Interpret** - Count the colonies. Confirmed coliforms are red or blue colonies with associated gas bubbles. Confirmed E. coli are blue colonies with associated gas bubbles. Results are fast and accurate.

ii) Water quality standard

As a first option, the projects shall meet the host country standards for treated water quality. Where national standards are not available, the project shall meet WHO standard of less than 1 Colony Forming Unit (CFU) of E. coli /100 ml. The 90/10 precision rule shall be followed in calculating the sample size required for testing water quality. The project technology entails water filters which is a 'point of use' technology. Therefore, the quality testing shall be done for samples taken at the water outlet.

The water quality testing results have complied with the host country standard for treated water quality of less than 1 Colony Forming Unit (CFU) of E. coli /100 ml. None of the households had water with detectable e. Coli as summarised in the results below.

Households water quality testing analysis

Table 35: Households WQT results

User/Non-user analysis based on WQT tests		
	Result	percentage
Users	70	100%
Non-users	0	0%
Total	70	100%

Source: GS11169 Water Quality Testing Results tab HHF WQT results

Institutions water quality testing analysis

Table 36: Institutions WQT results

User/Non-user analysis based on WQT tests		
	Result	percentage
Users	30	100%
Non-users	0	0%
Total	30	100%

Source: GS11169 Water Quality Testing Results tab IHF WQT results

Community filters water quality testing analysis

Table 37: Community filters WQT results

User/Non-user analysis based on water quality		
	Value	percentage
Users	8	100%
Non-users	0	0%
Total	8	100%

Source: GS11169 CF WQT Survey tab IHF Water quality testing

The PP also conducted water quality testing based on the qualitative criteria outlined by the Gold Standard. The assessment covered the following aspects:

- Reported incidence of water borne disease in child/members in last 30 days
- User satisfaction
- Level of hygiene including but limited to Hand washing & Safe storage practices being followed.

The results from this assessment are as follows:

Households Water Quality Analysis

Table 38: Qualitative WQT assessment for households

WATER QUALITY SURVEY ANALYSIS			
		Value	Percentage
Incidence of water borne disease	Yes	0	0%
	no	70	100%
	Total	70	100%
Level of satisfaction	High	68	97%
	Moderate	2	3%
	Low	0	0%
	Total	70	100%
Hygiene practices	Yes	70	100.0%
	no	0	0.0%
	Total	70	100%

Source: GS11169 Water Quality Testing Results tab HHF WQT responses

Institutions water quality analysis

Table 39: Qualitative WQT assessment for institutions

WATER QUALITY SURVEY ANALYSIS			
		Value	Percentage
Incidence of water borne disease	Yes	0	0%
	no	30	100%
	Total	30	100%
Level of satisfaction	High	28	93%
	Moderate	2	7%
	Low	0	0%
	Total	30	100%
Hygiene practices	Yes	30	100.0%
	no	0	0.0%
	Total	30	100%

Source: GS11169 Water Quality Testing Results tab IHF WQT responses

Community filters water quality analysis

Table 40: Qualitative WQT assessment for community filters

WATER QUALITY SURVEY ANALYSIS			
		Value	Percentage
Incidence of water borne disease	Yes	0	0%
	no	8	100%
	Total	8	100%
Level of satisfaction	High	8	100%
	Moderate	0	0%
	Low	0	0%
	Total	8	100%
Hygiene practices	Yes	8	100.0%
	no	0	0.0%
	Total	8	100%

Source: GS11169 CF WQT Survey tab IHF Water quality testing

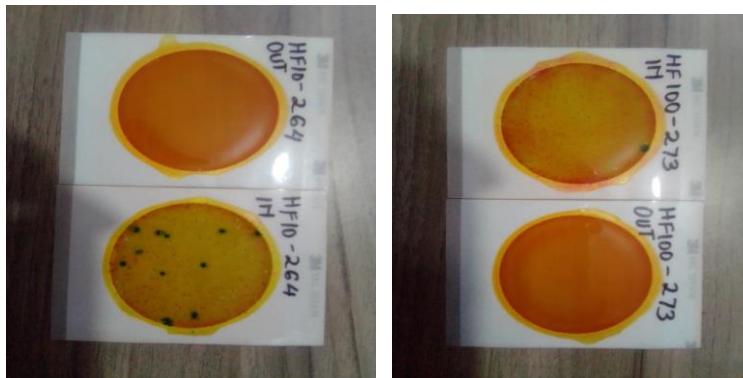


Figure 14: Water quality photos for filter numbers HF10-264 and HF100-273 respectively

e) Leakage Assessment

The core of the methodology prescribes possible parameters that should be evaluated for leakage emissions.

a) *The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.*

Traditional fuel wood consumption is mainly restricted to the domestic setting it is highly unlikely that fuel wood consumption in regions outside the target area will increase.

b) *The non-renewable biomass or fossil fuels saved under the project activity are used by non- project users who previously used lower emitting energy sources.*

Fuel wood previously used was sourced or obtained individually and domestic fuel wood consumption is a function of market forces of demand and supply. There is no indication that the fuel wood consumption would increase amongst non-project users who previously relied on lower-emitting sources.

c) *The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.*

The project will hence have minimal aggregate effect on the total NRB fraction in the country in comparison to other CDM or VER project activities baseline scenario.

d) *The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.*

In the project area, the average evening temperature is about 22°C and people do not use any space heating technology. Therefore, there is no loss of space heating because of filtering water. The project only displaces the practice of boiling water and households can still use the baseline stoves for cooking. Hence, there is no leakage as end-users do not light fires for space heating.

e) *By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.*

The BioSand Filter and the Hollow fibre membrane filters are zero energy technologies that will provide an alternative to boiling water using fuel wood. Most of the households targeted boil water using fuel wood. There is no envisioned scenario where these consumers will substitute fuel wood to a technology with lower emissions because of promotion of the project technologies.

In summary, Leakage risks of the project activity deemed very low and can be ignored. Leakage emissions (LE_y)=0 tCO₂e. Leakage investigation will be conducted every two years. The parameters of leakage emissions are included in the monitoring plan.

F) Non-Renewable Biomass Assessment Update

The NRB assessment update will not occur for the project given its fixed baseline and the fact that the project scale is not large enough to affect the national NRB fraction.

G) Water Consumption Field Tests

The water consumption field test (WCFT) measures project-supplied clean water consumption volumes. The WCFT is conducted with end user's representative of the project scenario target population and currently using the project technology. The WCFT must be conducted over 3 days, not including weekends, and averaged value (l/person/day) value should be determined after excluding outliers. It must be made explicit to the households/institutions that they must behave and consume water normally, reflecting typical daily water consumption pattern. The Minimum sample size for HWT is 30. In case of IWT, the minimum sample size shall be determined considering the project technology type and in line with the sampling approach applied. The WCFT exercise will be used to measure the parameter:

- QPW,p: Volume of drinking water per person per day for premises type p

The value obtained from the WCFT shall be capped at 5.5 L/person per day.

The WCFT will also record on a day-to-day basis the number of people living in the end user's household to determine the average number of people living in a household.

3. Quality Assurance and Quality Control

The project proponent commits to maintain accurate and transparent record keeping, monitoring and evaluation. All supporting documentation and records for the project must be easily accessible for spot checking and cross referencing by a third party; these records will be kept by ClimateCare. Contact information in the total sales record shall allow a project auditor to easily contact and visit end users. An auditor must also be able to cross reference pertinent project documentation, which must include archives such as production records (e.g., materials purchases, internal logs), financial accounts and sales records, as well as wholesale customer invoices, observations of retailer activities and sales performance. The PP shall perform statistical analysis on the data collated using Microsoft excel in an automated tool that will use standard deviation and mean values to determine outliers.

Survey and Testing Procedure

ACI shall ensure that all tests and surveys are:

1. Transparent and can easily be replicated,
2. Evidently conservative,
3. Sample is randomly selected so as to not introduce a material bias.
4. The impact of daily and seasonal variations on the expected average fuel consumption savings is accounted for.

Monitoring structure

The overall monitoring structure of the project is as indicated in the figure below. The Chief operations officer will establish the Monitoring, evaluation and learning team which will coordinate the monitoring of the project within the project boundary to facilitate data collection and recording. The monitoring team will also be responsible for planning, implementing, and monitoring WASH campaigns for the project. The team will be carefully selected and trained they have the required skills and expertise to perform the monitoring task.

Responsibility of each group is summarized as follows:

Sales team: headed by the regional sales coordinators will be responsible making sales and maintaining the sales database. The project proponent must maintain an accurate and complete electronic database for enabling the unique identification of the project water filters. It is envisaged that the database will include at least the following information:

- Date of sale
- Geographic area of sale
- Model/type of project technology sold and unique water purifier serial number
- Mode of use: commercial, institutional, domestic, and other
- Number of water filters sold
- Name, telephone number and address.

Monitoring, evaluation and learning team: charged with the duty of planning and carrying out project survey, usage survey, water quality survey tests, and jointly work with the WASH managers in organizing WASH/hygiene campaigns.

Monitoring, evaluation and learning manager: to conduct data verification, recording and filing and managing all the project documents.

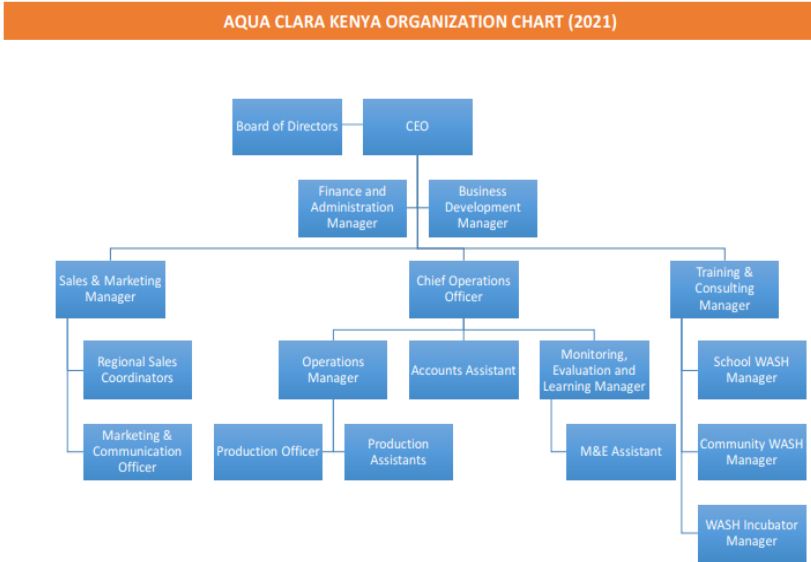


Figure 15: Aqua Clara Organisational Structure

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project.

01/05/2020 (The day the first filter was sold under the project).

C.1.2 Expected operational lifetime of project

20 years

The BioSand filters have a lifespan of 10+ years, while the hollow fibre filters have a lifespan of 5+ years with proper use and maintenance. After the 5 years, the PD shall make plans to replace the membranes of the filters that will have worn out so that the households and institutions can continue using the filters for till the end of the crediting period and longer.

C.2. Crediting period of project

C.2.1 Start date of crediting period.

09/02/2022 (start date for the design certification) or the end date of design review whichever falls latter. .

C.2.2 Total length of crediting period

15 years (5 years renewable twice)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 2	Project shall monitor the number of people employed by the project including their gender and income to ensure equal opportunities are given to both men and women, and that equal pay for equal value of work done is maintained.
Principle 6.1	The PP shall monitor the number of people employed by the project to ensure compliance with the national laws.
Principle 6.1	Employment records shall document the ages of the persons employed by the project. These shall be provided as supporting evidence to ascertain no children are employed by the project.
Principle 6.1	The training records for the trainings conducted on filter installation and maintenance shall be provided as supporting evidence.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	In order to ensure gender sensitive design and implementation, the project shall address any gender inequalities and gender-related risks that may be identified in project gender analysis. Gender-sensitive approaches in stakeholder consultation will also be ensured by promoting information sharing equitably with women and men stakeholders, which will be made
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	available and presented in an accessible format across all stakeholder groups including those more marginalized. The approach will also include opportunities for stakeholders to share information in a two-way exchange, give regular feedback during implementation and ensure their views and priorities are incorporated in project design and implementation. The project has further provided quantifiable 'easy-to-measure' indicators potentially aligned to the related SDG goals that measure gender-related gaps and risks for the project. Efforts shall also be geared towards innovatively increasing women's empowerment opportunities.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Government of Kenya adopted the National Policy on Gender and Development in 2000. The policy spells out a policy approach of Gender mainstreaming and empowerment of women and clearly states that it is the right of women, men, girls and boys to participate in and benefit equally from the development process³⁰. The plan of action (2008 – 2012) to implement the national policy on gender and development with an objective of

³⁰ <http://www1.uneca.org/Portals/ngm/Documents/GenderPolicy.pdf>

	facilitating the mainstreaming of gender concerns in all areas of the development process and to provide a basis for stakeholders to initiate programmes to promote gender equality in the country ³¹ . The project will provide job opportunities during its overall operation which will benefit both men and women.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No

³¹ http://www1.uneca.org/Portals/ngm/Documents/Gender_Mainstreaming%20Action%20Plan%20final%20-Apr08.pdf

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the physical consultation meeting and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

Below is a summary of the comments raised by the stakeholders during the meeting:

Gend er of Stak ehol der	Stakeholder comment	Was comment taken into account (Yes/ No)?	Explanation (Why? How?)
M	What is the cost of the filter?	Yes	HF15P costs KES. 3,700, the HF10P costs KES. 5,500 and Institutional Filter costs KES. 20,000.
M	How does the filter work?	Yes	ACK filters use the membrane filtration method to treat water by allowing the passage of water molecules and blocking any contaminants. Trapped contaminants are then removed through the backwashing process.
M	Does the filter require cleaning?	Yes	The membrane is cleaned using a syringe, by pulling out the water from the jar, a process known as backwashing. Also, the safe water storage container should be cleaned with soap at least once a week to ensure that the filtered water remains clean. However, the raw water container can be cleaned with clean water only to prevent any soap residues ending up in the membrane as this may lead to blockage.
M	At what stage are the germs killed since no chemicals used?	Yes	ACK filters use the method of hollow membrane filtration to treat water by allowing the passage of water molecules and blocking any contaminants. Trapped contaminants are then removed through the

			backwashing process, hence no chemicals used in the water treatment process.
M	Do all ACK filters use the membrane filtration method of treatment?	Yes	All ACK hollow fibre filters use the membrane filtration technology. However, the BioSand filters uses the sand filtration method.
M	Can one use any source of water for the filter?	Yes	ACK's method of filtration, with proper maintenance and handling of the filter, allows one to treat water from any source.
M	Are there records of sales?	Yes.	Upon sale of any ACK product, a sales receipt is issued to the client and a copy of the receipt is filed. ACK has also generated a sales database where all details of customers are entered to facilitate the monitoring process and end-user follow up.
M	Are there records to show the decline of water-borne diseases after the distribution of the filters?	Yes.	There are no official records to show the decline of water-borne disease after the distribution of the filter in all project areas. However, the monitoring process has so far indicated positive impacts from end users in terms of improved health.
F	Can water with chlorine affect the membrane?	Yes.	Chlorine in water does not affect the membrane.
F	Is it possible an office in Borabu to reach more people?	Yes.	ACK has agents in different regions to facilitate communication and distribution of products.
M	Can the water be tested before and after filtration to check the effectiveness of the filter?	Yes	The water quality monitoring of the filter involves testing of raw

			water and filtered water for e.Coli.
M	Can water tests results per region be shared for people to be aware of the status of their water and encourage more people to buy ACK filters?	Yes	More collaboration was encouraged between ACK and partners to facilitate the water testing process.
M	Several water sources are highly polluted because of their location, for instance near toilets, it is advisable that communities are sensitized and made aware of their water quality to take corrective action.	Yes	Water testing can be costly to an organization; hence they have prioritised on testing on E. coli and coliforms bacteria, with regard to national and international standards of drinking water.
M	A member suggested that a stream in Rigoma that was a water source to many community members was heavily polluted. He asked if ACK could conduct a water test and enlighten community members on the water status.	Yes	Water testing can be costly to an organization; hence they have prioritised on testing on E. coli and coliforms bacteria, with regard to international standards of drinking water.
F	A member suggested that PHOs could liaise with local chiefs and community leaders since they understand more about the people and the region and could give priorities to people whose water sources are very unhealthy or have previously suffered from water-borne diseases. for water testing. Liaising with local community leaders would also facilitate data dissemination process with community members.	Yes.	CK will explore better ways of engaging with the community leaders to ensure maximum benefits are achieved. Key among the activities will be undertaking hygiene campaigns.
F	On behalf of all stakeholders, a member stated that the community members have greatly benefitted from the filters through improved health and capacity building.	Yes	ACK appreciated the positive feedback and committed to continue supporting local communities.

From the comments, no negative issues were raised that require mitigations.

E.2 Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Aqua Clara Kenya Office Rigoma, Nyamira County
GS Contact (mandatory)	help@goldstandard.org info@goldstandard.org
Telephone access (optional)	ACK Office phone numbers (+254) 0716 406687 (+254) 0114175815 (+254) 0114170853
Internet/email access	infokenya@aquaclara.org
Nominated Independent Mediator (optional)	Borabu Sub-County Mr. David Nyaramba Phone number: 0717758559 Rigoma location Mr. Richard Moses Phone number: (+254) 0711275089 Narok County Ms. Eunice Punyua Phone number: (+254) 0729436429 Tom Tumpuya Phone number: (+254) 0721399588

Stakeholder feedback round

The stakeholder feedback round (SFR) commenced on 24/08/2021 and covered a period of 60 days till 24/10/2021. The consultation was conducted to solicit more views from the participants and ensure that any comments/concerns brought up during the LSC meetings are effectively addressed.

As agreed during the LSC meeting, the SFR process was conducted through emails as this was deemed appropriate for majority of the stakeholders who attended the meeting. The project documents were also printed and availed for review at the Aqua Clara office in Rigoma, Nyamira County. The invitations to take part in the SFR were

sent out to all stakeholders who had been invited to the meeting and included also the meeting attendees. The SFR took a period of two months (60 days), and participants were encouraged to take part in the process to ensure all their comments are addressed.

The documents shared with the stakeholders during the process included:

- The non-technical summary for the project (English and Swahili versions)
- Preliminary review feedback final form
- Project Design Documents
- Stakeholder consultation report

No comments were received from this consultation.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights	No	The project is not complicit in Human rights abuses. The project activity shall respect human rights during its operation and there shall be minimal possibility of human right abuses.	No mitigation required.
2. The Project shall not discriminate with regards to participation and inclusion	No	The project activity shall adopt a non-discrimination approach to ensure equal inclusion and participation of men and women in the opportunities that arise.	No mitigation required.

Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women.	No	The project does not directly or indirectly lead to gender-based discrimination, nor does it contribute to adverse impacts on gender equality. The project gives equal opportunities to both men and women by involving them in the project and giving them access to the project.	No mitigation required.
2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work.	Yes	The project seeks to promotes gender equality and the empowerment of women. The project neither contributes to discrimination against women nor does it reinforce gender-based discrimination and/or inequalities.	Project shall monitor the number of people employed by the project including their gender and income to ensure equal opportunities are given to both men and women, and that equal pay for equal value of work done is maintained.
3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks.	No	The Government of Kenya adopted the National Policy on Gender and Development in 2000. The policy spells out a policy approach of Gender mainstreaming and empowerment of women and	No mitigation required.

4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	clearly states that it is the right of women, men, girls and boys to participate in and benefit equally from the development process. In compliance with this policy, the project activity shall seek to address any potential gender risks by ensuring equal participation of both men and women in the project. Expert stakeholder has not been included in the assessment.	No mitigation required
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	Using the water filters will provide safe water for consumption by households. This shall lead to reduced expenditure on medical bills and treatment of diseases associated with indoor air pollution and water borne disease. As a result, the project shall contribute	No mitigation required

		towards reducing community exposure to increased health risks associated with unsafe water and indoor pollution.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project does not involve and is not complicit in the alteration, damage or removal of any critical historic, cultural, traditional, or religious heritage. The project is a water filtration program.	No mitigation required
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not involve and is not complicit in involuntary resettlement or displacement of locals. The proposed project activity involves distribution and use of water filters in households and shall not lead to involuntary resettlement or displacement.	No mitigation required
>>			
Principle 4.3 Land Tenure and Other Rights			

a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No	The proposed project activity involves distribution and installation of water filters. There shall be no change in land tenure arrangements or involuntary resettlement because of implementation of the project activity.	No mitigation required
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project recognises and respects the rights of indigenous peoples and shall seek to respect, protect, and conserve the cultural, intellectual, religious and spiritual property of indigenous people. The Project shall identify any communities of Indigenous People within the Project area who may be affected directly or indirectly by the Project. However, the proposed project activity involves	No mitigation required
>>			

		distribution of water filters and shall have no impacts on indigenous people.	
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The project does not involve and is not complicit in corruption. The project activity shall not engage in, contribute to or reinforce corruption of any kind and all activities will be done in a transparent manner.	No mitigation required
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions. 2. Workers shall be able to establish and join labour organisations.	Yes	Yes, the project does not involve and is not complicit in any form of forced or compulsory labour. Employment shall be in compliance with laid out national and international laws and procedures. Participation into the project activity shall be voluntary.	The PP shall monitor the number of people employed by the project to ensure compliance with the national laws.

3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave.	Yes	The project developer shall ensure proper documentation for all employees including the working agreements.	Contracts will be signed by the permanent staff employed by the project detailing the working agreements as required.
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4.	No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)	Yes	The project shall ensure that no children (below 18years of age) are employed by the project.	Employment records shall document the ages of the persons employed by the project. These shall be provided as supporting evidence to ascertain no children are employed by the project.
5.	The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures	Yes	The project shall conduct relevant trainings to the filter installers contracted by the project to conduct filter installations. The filter assembly and installation are simple procedures that does not require sophisticated equipment. No major accidents or incidences can be anticipated from this process.	The training records for the trainings conducted on filter installation and maintenance shall be provided as supporting evidence.
Principle 6.2 Negative Economic Consequences				
1.	Does the project cause negative economic consequences during and after project implementation?	No	The project will not cause negative economic consequences but rather promote creation of jobs in the targeted areas which is a	No mitigation required

>>		positive economic consequence.	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	There are no anticipated project emissions from the project activity given that the project introduces zero energy water filters. These technologies do not use any non-renewable biomass and hence displace the practice of boiling water completely. By reducing the consumption of unsustainably harvested fuel wood GHG emissions are reduced.	No mitigation required
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The water filters operate using a combination of biological and physical processes that does not involve use of energy supply. The project will therefore not use energy from a local grid or power supply.	No mitigation required
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			

Will the Project affect the natural or pre-existing pattern of watercourses, groundwater and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project activity consists of zero energy Point of Use and Point of Entry water treatment at household levels, that does not impact on natural water patterns and flow.	No mitigation required
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	Minimization of deforestation in the project boundary as a result of the project will lead to reduction in the occurrence and severity of soil erosion. The project area is not susceptible to excessive erosion or water body instability and the proposed project activity shall not negatively affect soil conditions in the project scenario.	No mitigation required
>>			
Principle 9.1 Landscape Modification and Soil			

Does the Project involve the use of land and soil for production of crops or other products?	No	The project is a water filtration project that aims at providing and does not involve production of crops.	No mitigation required
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project does not create nor increase vulnerability to natural disasters. Using the water filters displaces the use of fuelwood for boiling water in the baseline which leads to reduced fuelwood, and this has a positive impact in reducing deforestation. This will reduce the susceptibility of project areas to natural disasters such as landslides, flooding, or soil erosion.	No mitigation required
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or	No	The project activity is a water filtration project and does not involve the use of GMOs in its operations.	No mitigation required

take place in facilities or farms that include GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The water filters use a combination of biological, and physical processes to treat water. No chemicals or hazardous materials are used in the filtration process. The project has therefore no potential of releasing pollutants to the environment.	No mitigation required
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The assembly, installation and use of the water filters involve light manual works and uses locally available materials such as sand. The process does not involve use of any chemicals or other hazardous materials.	No mitigation required
>>			
Principle 9.6 Pesticides & Fertilisers			

Will the Project involve the application of pesticides and/or fertilisers?	No	The project is a water filtration project that does not use pesticides nor fertilizers.	No mitigation required
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project activity does not entail harvesting of forests and actually it helps to improve natural habitats by reducing the demand for charcoal and wood.	No mitigation required
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project is a water filtration project and does not affect crop production, quality, quantity and availability of the same in any way.	No mitigation required
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project is a water filtration project that aims at providing affordable, clean, efficient, and easy-to-use alternative for water	No mitigation required
>>			

		purification. It therefore does not involve animal husbandry practices.	
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project does not involve and is not complicit in significant conversion or degradation of critical natural habitats, including those that are (a) legally protected, (b) officially proposed for protection, (c) identified by authoritative sources for their high conservation value, or (d) recognized as protected by traditional local communities. The proposed project activity target area shall be households.	No mitigation required
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	The project shall not lead to the reduction or negative impact of any recognised endangered, vulnerable or critically endangered species. The project activity is a water	No mitigation required.

b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		filtration project which will entail installation of filters at households. The project activity shall not impact on any habitats of endangered species since it entails installation of filters at households.	No mitigation required
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APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

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APPENDIX 3- LUF ADDITIONAL INFORMATION
 N/A

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change [Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption